Lecture # 01

## Operation Research:

What is Operation Research?

Operation Research (O.R) is an art of wining wars without actual fighting.

(Arthur Clark)

O.R is a scientific approach to the problems.

(H.M. Wagner)

## Linear Programming:

Linear programming is a mathematical technique for determining the optimal solution and obtaining a particular objective when there are alternative uses of resources. The objective may be cost minimization or profit maximization.

The word linear means that the relationships are represented by straight line ax+by = c

The word programming that is concerned with optimal allocation of limited resources.

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## Linear Function:

A linear function contains terms of which is composed of only a single variable raised to the power one. Linear functions are those whose graph is a straight line. e.g. 3x+2y = 7 (linear) , $3x^{3/2}$+2y = 7 (Non-linear)

## Objective Function:

It is a linear function of decision variables. z = $x_1$+$x_2$ is the most typical form of objective functions are maximize f(x) or minimize f(x).

Decision Variable وہ ہوتے ہیں جو decide کرتے ہیں کہ فنکشن کہاں پر maximum ہے اور کہاں پر minimum ہے۔

## Constraints:

These are the linear equation arising out of practical limitations. The mathematical forms of constraints are f(x) ≤ b or f(x) ≥ b or f(x) = b

**Feasible Solution:** A non-negative solution which satisfies all the constraints is known as feasible solution. The region comprising all feasible solutions is referred to as feasible region.

## Optimal Solution:

The solution where the objective function is maximize or minimize is known as optimal solution.

## General linear programming problem:

Consider the following optimize

$$z = c_1x_1 + c_2x_2 + ..... + c_nx_n \quad subject\ to$$

$$a_{11}x_1 + a_{12}x_2 + ..... + a_{1n}x_n \quad which\ is\ (\leq, \geq, =) b_1$$

$$a_{21}x_1 + a_{22}x_2 + ..... + a_{2n}x_n \quad which\ is\ (\leq, \geq, =) b_2$$

. . .

. . .

. . .

$$a_{m1}x_1 + a_{m2}x_2 + ..... + a_{mn}x_n \quad which\ is\ (\leq, \geq, =) b_m$$

$x_1, x_2, ...., x_n \geq 0$ In summation form the above problem can be written as

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$$z = \sum_{i=1}^{n} c_j x_j \quad subject\ to$$

$$\sum_{j=1}^{n} a_{ij} x_j \ (\leq, \geq, =) \ b_i, i = 1, 2, ...., m$$

$$and\ \ x_j \geq 0 \quad j = 1, 2, ...., n$$

$c_{j's}, a_{ij's}, b_{i's}$ are decision variables.

## Graphical Solution of Linear Programming problem:

**Question:** $2x \leq 4$

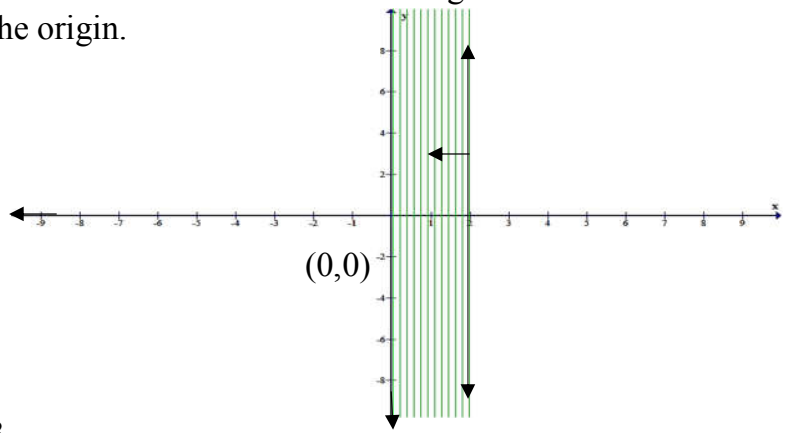
Solution: $x \leq 2$

Associated equation is

$x = 2$

At (0,0) $\Rightarrow$ $0 \leq 4$

If at (0,0) the equation is true then solution is toward origin & if false the solution is away from the origin.




**Question:** $2x - 2y \le 3$

Solution: Associated equation is

$2x - 2y = 3$

At $y = 0$ $\quad 2x = 3 \quad \Rightarrow \quad x = \frac{3}{2} \quad \Rightarrow \quad \left(\frac{3}{2}, 0\right)$

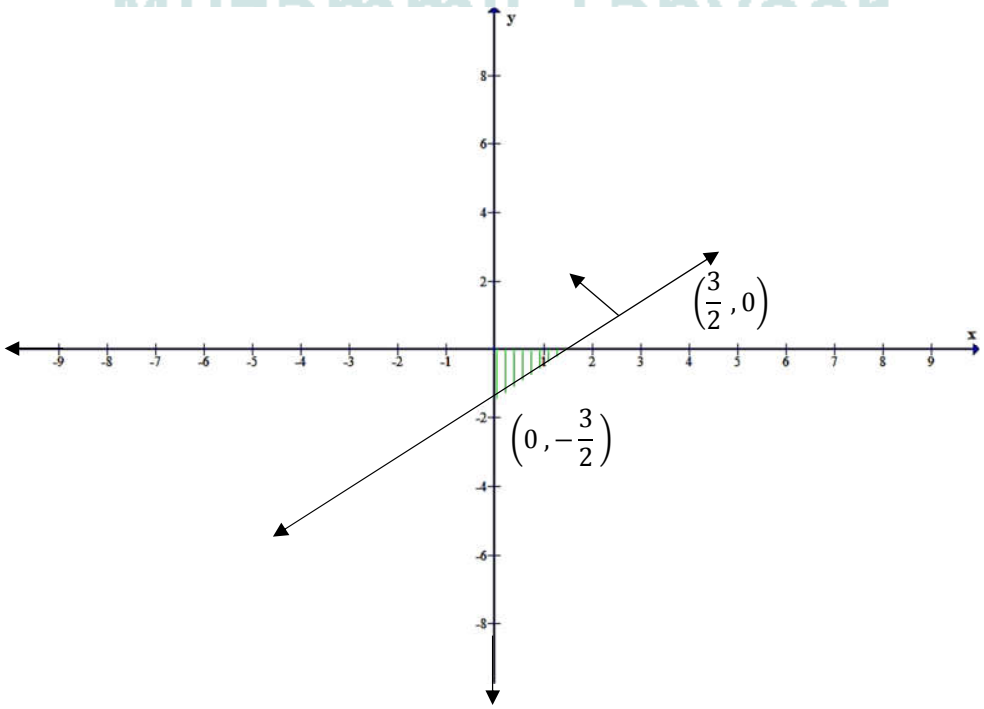
At $x = 0$ $\quad -2y = 3 \quad \Rightarrow \quad y = -\frac{3}{2} \quad \Rightarrow \quad \left(0, -\frac{3}{2}\right)$

At (0,0)

$0 - 0 \le 3 \quad \Rightarrow \quad 0 \le 3$ true

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Then solution is toward origin.



## Corner Point:

A point of solution region where two of its boundary line are intersect is called corner point or vertex of solution region.

**Example:** Maximize $z = x_1 + 3x_2$ Subject to $x_1 - x_2 \leq 5$

$x_1 \geq 2$ , $x_2 \leq 5$

$x_1, x_2 \geq 0$

Solution: Given that $x_1 - x_2 \leq 5$

Associated Equation is

$x_1 - x_2 = 5$

At $x_2 = 0 \Rightarrow x_1 = 5 \Rightarrow (5,0)$

At $x_1 = 0 \Rightarrow -x_1 = 5$ *or* $x_1 = -5 \Rightarrow (0,-5)$

And $x_1 \geq 2$

Associated equation is

$x_1 = 2 \Rightarrow (2,0)$

*And* $x_2 \leq 5$

Associated equation is

$x_2 = 5 \Rightarrow (0,5)$

At (0,0) for $x_1 - x_2 \leq 5$

$0 - 0 \leq 5 \Rightarrow 0 \leq 5$ *true*

Solution toward origin

For $x_1 \geq 2$

$0 \geq 2$ *false*

Solution away from origin

For $x_2 \leq 5$

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$0 \le 5$ *true* Solution toward origin

For corner point B $x_1 = 2, x_2 = 5 \Rightarrow (2,5)$

For corner point A (intersection of line $x_1 - x_2 = 5$ & $x_2 = 5$ )

$x_1 - x_2 = 5 \Rightarrow x_2 = 5$

$x_1 - 5 = 5 \Rightarrow x_1 = 10 \Rightarrow (10,5)$

For maximize $z = x_1 + 3x_2$

Put (2,0) $\Rightarrow$ z = 2

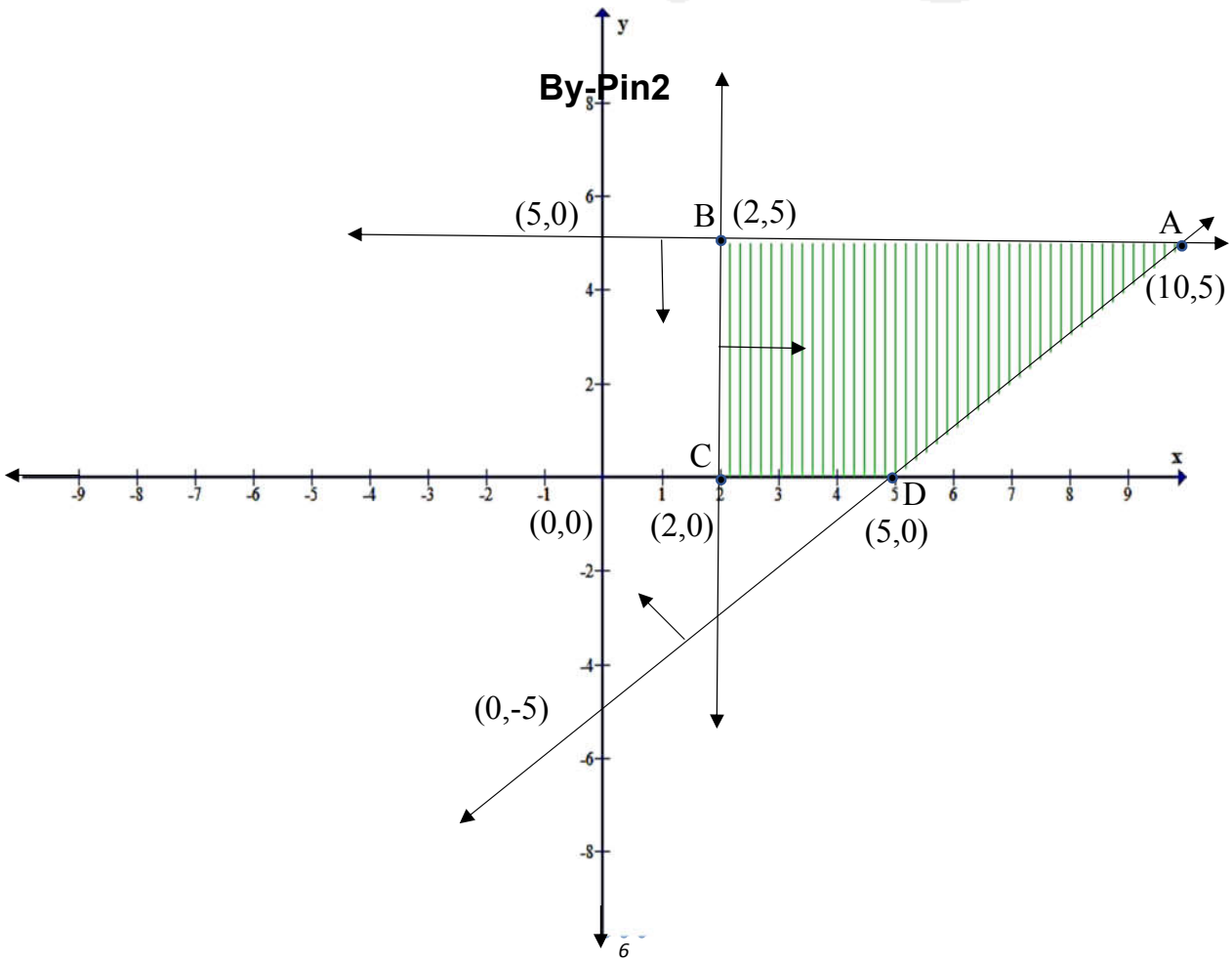
Put (5,0) $\Rightarrow$ z = 5

Put (2,5) $\Rightarrow$ z = 2 +15 =14

Put (10,5) $\Rightarrow$ z = 10+15 =25

So maximum at (10,5)

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**Example:** Find the equation from graph.

Solution: We know

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

*For* $(0,3)$ & $(4,0)$

$$y - 3 = \frac{0 - 3}{4 - 0}(x - 0)$$

$$y - 3 = \frac{-3}{4}x$$

$$4(y - 3) = -3x$$

$$4y - 12 = -3x$$

$$3x + 4y = 12$$

*For* $(0,-2)$ & $(1,0)$

$$y + 2 = \frac{0 + 2}{1 - 0}(x - 0)$$

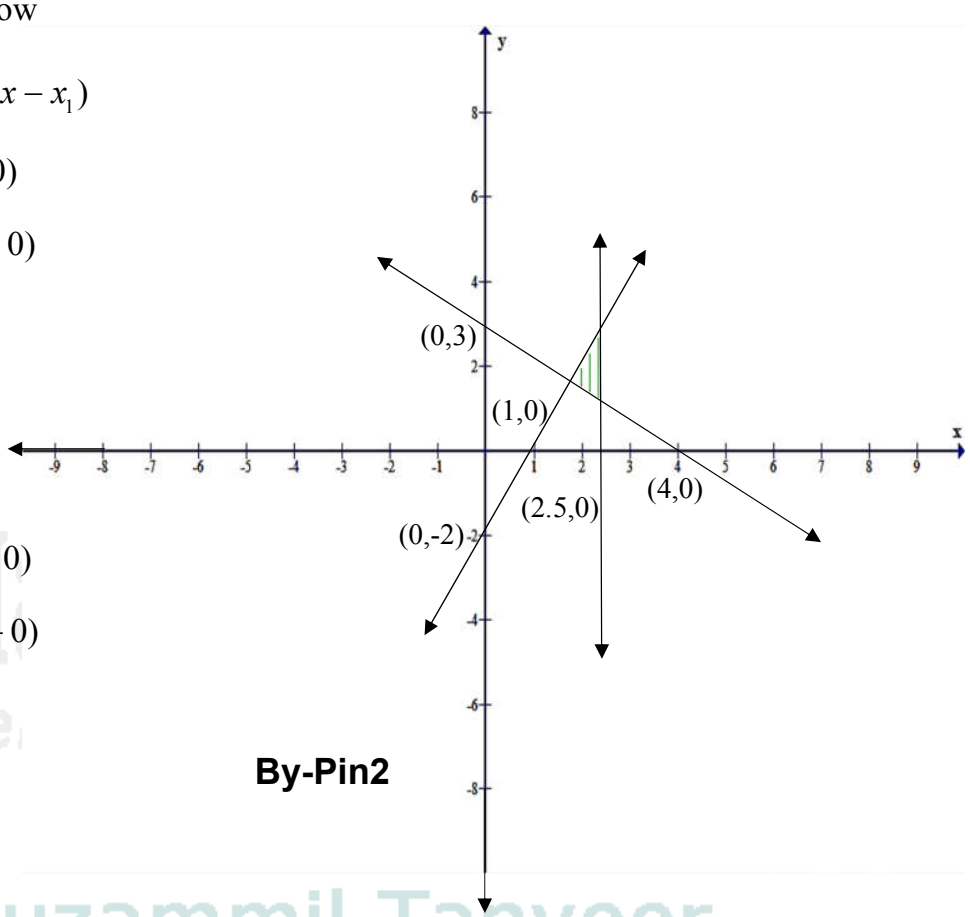
$$y + 2 = \frac{2}{1}(x)$$

$$y + 2 = 2x$$

$$2x - y = 2$$

*For* $(2.5,0)$

$$\Rightarrow x = 2.5$$




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Now (0,3) & (4,0) has solution away from origin. So

$$3x+4y \geq 12$$

And (0,–2) & (1,0) has solution away from origin. So

$$2x-y \geq 2$$

Also (2.5, 0) has solution toward origin. So

$$x \leq 2.5$$

Lecture # 2

## Special Cases in Graphical:

(i) Multiple Optimal Solutions:

More than one solution with the same optimal value of the objective function.

## Example:

*Maximize* $z = 2x_1 + x_2$

*Subject to*

$x - 2y \le 6$

$2x + y \le 2$

$x, y \ge 0$

*Solution* :

*Given that*

$x - 2y \le 6$

*Associated equation is*

$x - 2y = 6$

$At \quad y = 0 \Rightarrow x = 5 \Rightarrow (6,0)$

$At \quad x = 0 \Rightarrow -2y = 6 \Rightarrow x_2 = -3$

$\Rightarrow (0,-3)$

$And \ x \ge 0$

*Associated equation is*

$2x + y = 2$

$At \ y = 0 \Rightarrow x = 1 \Rightarrow (1,0)$

$At \ x = 0 \Rightarrow y = 2 \Rightarrow (0,2)$

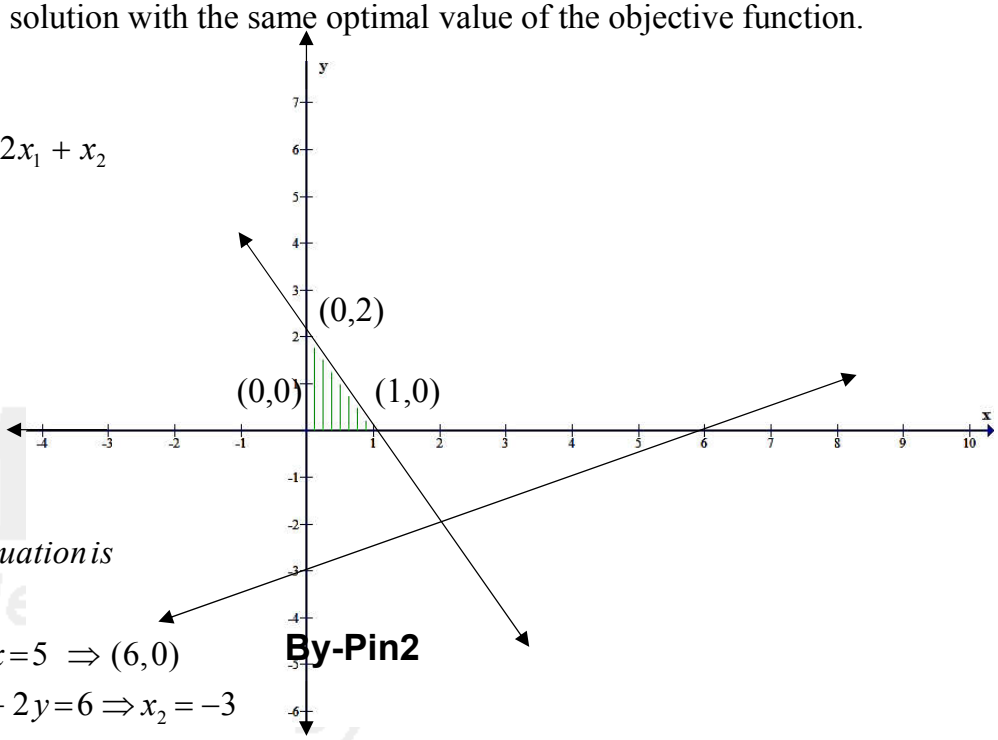
$And \ y \ge 0$

$z = 2x_1 + x_2$

$at \ (0,0) \Rightarrow z = 0$

$at (1,0) \Rightarrow z = 2$

$at \ (0,2) \Rightarrow z = 2$




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## Infeasible Region:

In some case there is no feasible solution area that is there are no points which satisfy all the constraints (inequalities).

$z = 2x_1 + x_2$

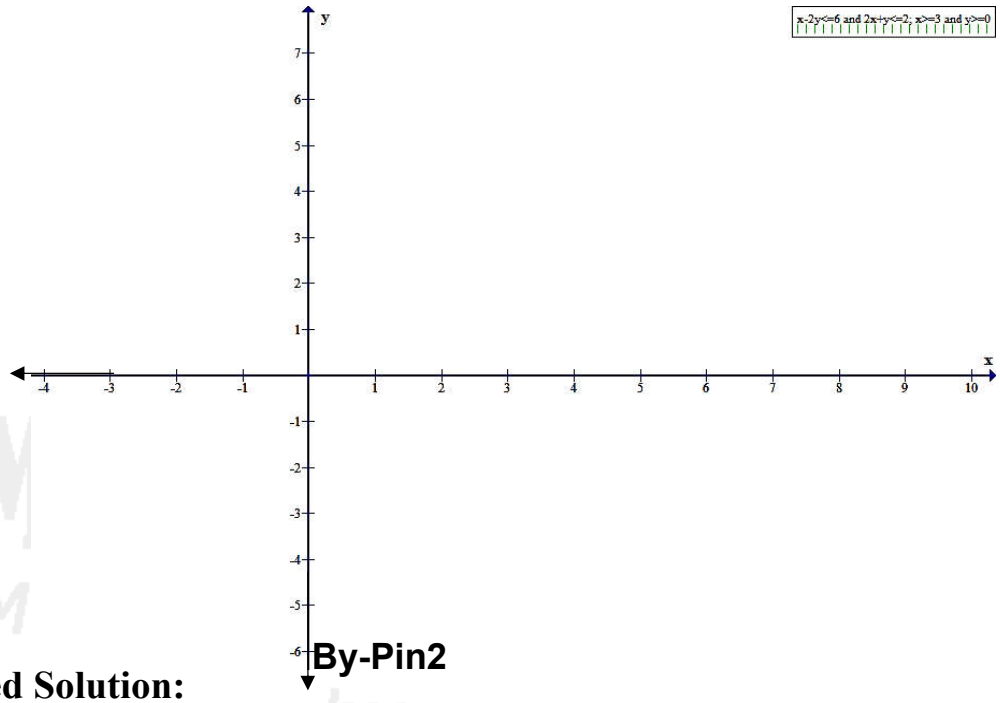
*Subject to*

$x - 2y \le 6$

$2x + y \le 2$

$x \ge 3$

$x, y \ge 0$




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## Unbounded Solution:

If the value of the objective function is increased indefinitely such solutions are called unbounded solution.

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**Example:** $x-2y \le 6$

$2x + y \ge 2$

$x , y \ge 0$

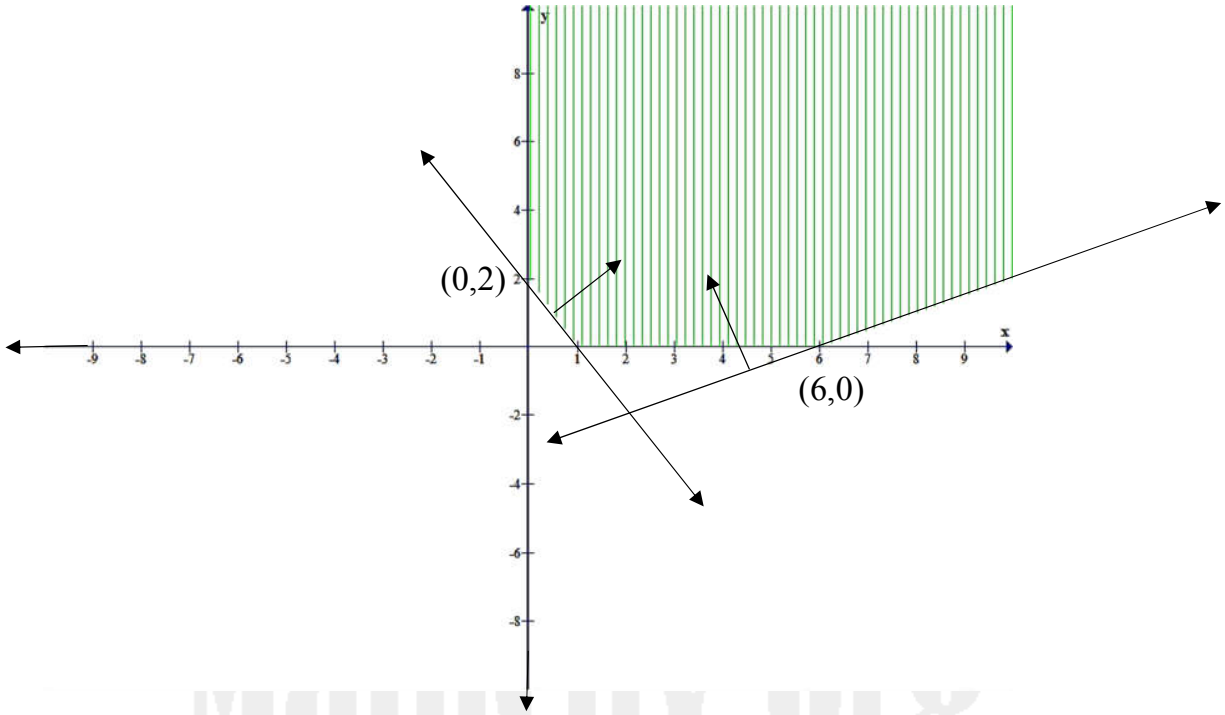
Solution: Given that

$x-2y \le 6$ , $2x + y \ge 2$

Associated Equation is

$x-2y = 6$ , $2x + y = 2$

| | | | | |
|---|---|---|---|---|
| $x = 6$ | (6,0) | , | $x = 1$ | (1,0) |
| $y = -3$ | (0,-3) | , | $y = 2$ | (0,2) |




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**Question:** A person requires 10,12 & 12 units chemical A, B & C respectively for his garden. A liquid product contains 5 ,2 & 3 units of A,B & C respectively per jar.A dry product contains 3,3 and 4 units of A, B and C respectively per carton. If the liquid product sells for Rs. 3 per jar & dry product sells for Rs. 2 per carton. How many of each should be purchase to minimize the cost and meet the requirment. Only formulate the above problem.

$Minimize\, z = 3x_1 + 2x_2$

$Subject\, to$

$5x_1 + 3x_2 \geq 10$ ..... $For\ \ A$

$2x_1 + 3x_2 \geq 12$ ..... $For\ \ B$

$3x_1 + 4x_2 \geq 12$ ..... $For\ \ C$

$x_1, x_2 \geq 0$

## Simplex Method for Solving Linear Programming Problem:

Drawback of simplex method that it solves only ≤ constraints.

## Slack Variable:

$$x_1 + x_2 \leq 3$$

To change above inequality, we add some variable that variable is called slack variable.

$$x_1 + x_2 + s_1 = 3$$

It is the variable that is added to the L.H.S of a less than or equal $\leq$ type constraints to convert inequality into equality.

## Surplus Variable:

It is a variable which is subtracted from the L.H.S of a greater than or equal to $\geq$ type constraints to convert inequality into equality.

e.g. $x_1 + x_2 \geq 9$

$$x_1 + x_2 - s_2 = 9$$

**Example:** Find maximum solution of the following problem by simplex method.

$$z = x_1 + x_2$$

Subject to $2x_1 + x_2 \leq 4$

$$x_1 + 2x_2 \leq 3$$

$$x_1, x_2 \geq 0$$

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Solution:

$$z = x_1 + x_2 + 0s_1 + 0s_2$$

Subject to $2x_1 + x_2 + s_1 = 4$

$$x_1 + 2x_2 + s_2 = 3$$

$$x_1, x_2, s_1, s_2 \geq 0$$

## Initial basic feasible solution:

We assume that nothing can be produced. Therefore the values of decision variable $x_1, x_2 = 0$ and also $z = 0$. So we left with unused capacities $s_1 = 4$ & $s_2 = 3$

Variables with non-zero values are called Basic variables and with zero values are called non-basic variables.

## Gauss Jordan method:

$$z - x_1 - x_2 - 0s_1 - 0s_2 = 0$$

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol |
|---|---|---|---|---|---|
| z | -1 | -1 | 0 | 0 | 0 |
| $s_1$ | (2) | 1 | 1 | 0 | 4 |
| $s_2$ | 1 | 2 | 0 | 1 | 3 |

Pivot element

## Entering Variable:

Most -ve value in z row (for maximization)

Most +ve value in z row (for minimization)

## Leaving Variable:

Minimum +ve ratio in ratio column (for both maximize and minimize)

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Entering Value

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol |
|---|---|---|---|---|---|
| z | -1 | -1 | 0 | 0 | 0 |
| $x_1$ | 1 | 1/2 | 1/2 | 0 | 2 |
| $s_2$ | 1 | 2 | 0 | 1 | 3 |

Leaving variable

Divide by 2 to make pivot element 1

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | |
|---|---|---|---|---|---|---|
| z | 0 | -1/2 | 1/2 | 0 | 2 | $z + x_1$ |
| $x_1$ | 1 | 1/2 | 1/2 | 0 | 2 | |
| $s_2$ | 0 | 3/2 | -1/2 | 1 | 1 | $s_2 - x_1$ |

Entering Value

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol |
|---|---|---|---|---|---|
| z | 0 | -1/2 | 1/2 | 0 | 2 |
| $x_1$ | 1 | 1/2 | 1/2 | 0 | 2 |
| $x_2$ | 0 | 1 | -1/3 | 2/3 | 2/3 |

Leaving variable $s_2$

Mult

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | |
|---|---|---|---|---|---|---|
| z | 0 | 0 | 1/3 | 1/3 | 7/3 | $z+\frac{1}{2}x_1$ |
| $x_1$ | 1 | 0 | 2/3 | -1/3 | 5/3 | |
| $x_2$ | 0 | 1 | -1/3 | 2/3 | 2/3 | $x_1-\frac{1}{2}x_2$ |

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Since all the value in z-row are non-negative. So, the solution obtained is optimal with

$$x_1=\frac{5}{3},\ x_2=\frac{2}{3},\ z=\frac{7}{3}$$

*and* $s_1=s_2=0$

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Lecture # 3

## Special Case in Simplex method:

There are two special case in simplex method

(i) Unbounded solution
(ii) Multiple optimal solution

### (i) Unbounded Solution:

**Question:** Find maximum solution of the following system of linear equation

$$z = 5x_1 + 4x_2$$

Subject to $x_1 \le 7$

$$x_1 - x_2 \le 8$$

$$x_1, x_2 \ge 0$$

Solution: $z = 5x_1 + 4x_2 + 0s_1 + 0s_2$

$$z - 5x_1 - 4x_2 - 0s_1 - 0s_2 = 0$$

$$x_1 + s_1 = 7$$

$$x_1 - x_2 + s_2 = 8$$

$$x_1, x_2, s_1, s_2 \ge 0$$

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| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | R/C |
|---|---|---|---|---|---|---|
| z | -5 | -4 | 0 | 0 | 0 | |
| $s_1$ | 1 | 0 | 1 | 0 | 7 | 7/1=7 |
| $s_2$ | 1 | -1 | 0 | 1 | 8 | 8/1=8 |

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | R/C | |
|---|---|---|---|---|---|---|---|
| z | 0 | -4 | 5 | 0 | 35 | | $z+5x_1$ |
| $x_1$ | 1 | 0 | 1 | 0 | 7 | $7/0=\infty$ | |
| $s_2$ | 0 | -1 | -1 | 1 | 1 | $-1/1=-1$ | $s_2-x_1$ |

Since there is no minimum positive value in Ratio column. So, it is not possible to proceed further with simplex method. This is the criteria for unbounded solution in simplex method.

## (ii) Multiple optimal solution:

The optimal solution may not be unique if the non-basic variable has a zero co-efficient in z row.

This implies that bringing the non-basic variable into the basic will neither increase nor decrease the value of the objective function. Thus, the problem has multiple optimal solution.

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**Example:** Maximize $z=2x_1+3x_2$

Subject to $6x_1+9x_2\le 100$

$2x_1+x_2\le 20$

$x_1,x_2\ge 0$

Solution: $z=2x_1+3x_2+0s_1+0s_2$

$z-2x_1-3x_2-0s_1-0s_2=0$

$6x_1+9x_2+s_1=100$

$2x_1-x_2+s_2=20$

$x_1,x_2,s_1,s_2\ge 0$

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | R/C |
|---|---|---|---|---|---|---|
| z | -2 | -3 | 0 | 0 | 0 | |
| $s_1$ | 6 | 9 | 1 | 0 | 100 | 100/9=11.11 |
| $s_2$ | 2 | 1 | 0 | 1 | 20 | 20/1=20 |

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | |
|---|---|---|---|---|---|---|
| z | -2 | -3 | 0 | 0 | 0 | |
| $x_2$ | $\frac{2}{3}$ | 1 | $\frac{1}{9}$ | 0 | $\frac{100}{9}$ | $\frac{1}{9}s_2$ |
| $s_2$ | 2 | 1 | 0 | 1 | 20 | |

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| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | |
|---|---|---|---|---|---|---|
| z | 0 | 0 | $\frac{1}{3}$ | 0 | $\frac{100}{3}$ | $z+3x_2$ |
| $x_2$ | $\frac{2}{3}$ | 1 | $\frac{1}{9}$ | 0 | $\frac{100}{9}$ | |
| $s_2$ | $\frac{4}{3}$ | 0 | $-\frac{1}{9}$ | 1 | $\frac{80}{9}$ | $s_2-x_2$ |

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$$x_1=0,\ x_2=\frac{100}{9},\ s_1=0,\ s_2=\frac{80}{9}, z=\frac{100}{3}$$

**For Finding multiple solution:**

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | R/C |
|---|---|---|---|---|---|---|
| z | 0 | 0 | $\frac{1}{3}$ | 0 | $\frac{100}{3}$ | |
| $x_2$ | $\frac{2}{3}$ | 1 | $\frac{1}{9}$ | 0 | $\frac{100}{9}$ | $\frac{\frac{100}{9}}{\frac{2}{3}}=\frac{50}{3}$ |
| $s_2$ | $\frac{4}{3}$ | 0 | $-\frac{1}{9}$ | 1 | $\frac{80}{9}$ | $\frac{\frac{80}{9}}{\frac{4}{3}}=\frac{20}{3}$ |

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| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | |
|---|---|---|---|---|---|---|
| z | 0 | 0 | $\frac{1}{3}$ | 0 | $\frac{100}{3}$ | |
| $x_2$ | $\frac{2}{3}$ | 1 | $\frac{1}{9}$ | 0 | $\frac{100}{9}$ | |
| $x_1$ | 1 | 0 | $-\frac{1}{12}$ | $\frac{3}{4}$ | $\frac{20}{3}$ | $\therefore \frac{3}{4}s_2$ |

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | |
|---|---|---|---|---|---|---|
| z | 0 | 0 | $\frac{1}{3}$ | 0 | $\frac{100}{3}$ | |
| $x_2$ | 0 | 1 | $\frac{1}{6}$ | $-\frac{1}{2}$ | $\frac{20}{3}$ | $x_2-\frac{2}{3}x_1$ |
| $x_1$ | 1 | 0 | $-\frac{1}{12}$ | $\frac{3}{4}$ | $\frac{20}{3}$ | |

$$x_1=\frac{20}{3},\ x_2=\frac{20}{3},\ s_1=0,\ s_2=0, z=\frac{100}{3}$$

Lecture # 04

## Degeneracy:

In some cases, there may be doubt in selecting the variable that should be introduced into the basic i.e. there is a tie between the ratio of two variables.

To resolve degeneracy, we select one of them arbitrary.

**Example:** Maximize $z = 3x_1 + 9x_2$

Subject to $x_1 + 4x_2 \le 8$ , $x_1 + 2x_2 \le 4$

$x_1, x_2 \ge 0$

Solution: $z = 3x_1 + 9x_2 + 0s_1 + 0s_2$

$z - 3x_1 - 9x_2 - 0s_1 - 0s_2 = 0$

$x_1 + 4x_2 + s_1 = 8$

$x_1 + 2x_2 + s_2 = 4$

$x_1, x_2, s_1, s_2 \ge 0$

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| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | R/C |
|---|---|---|---|---|---|---|
| z | -3 | -9 | 0 | 0 | 0 | |
| $s_1$ | 1 | 4 | 1 | 0 | 8 | $\frac{8}{4} = 2$ |
| $s_2$ | 1 | 2 | 0 | 1 | 4 | $\frac{4}{2} = 2$ |

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol | |
|---|---|---|---|---|---|---|
| z | $\frac{3}{2}$ $\frac{9}{2}$ | 0 | 0 | | 18 | $z + 9x_2$ |
| $s_1$ | -1 1 | 0 | 1 | -2 | 0 | $s_1 - 4x_2$ $\frac{1}{2}x_2$ |

$$x_1=0,\ x_2=2,\ s_1=0\ ,z=18$$

## Unrestricted Variable:

Sometime variables are unrestricted in sign (+,–,0). In all such cases the decision variables can be expressed as the difference between two non-negative variables.

For example, if $x_1$ is unrestricted in sign then we write $x_1=x_1^{'}-x_1^{''}$

**Example:** Maximize $z=2x_1+3x_2$

Subject to $-x_1+2x_2\le 4$

$x_1+x_2\le 6$

$x_1+3x_2\le 9$ where $x_1,x_2$ are unrestricted in sign

Solution: Here $x_1$ and $x_2$ both are unrestricted in sign so we put

$$x_1=x_1^{'}-x_1^{''}\ \ \&\ \ x_2=x_2^{'}-x_2^{''}$$

$$z=2x_1^{'}-2x_1^{''}+3x_2^{'}-3x_2^{''}+0s_1+0s_2+0s_3$$

$$z-2x_1^{'}+2x_1^{''}-3x_2^{'}+3x_2^{''}-0s_1-0s_2-0s_3=0$$

$$-x_1^{'}-x_1^{''}+2x_2^{'}-2x_2^{''}+s_1=4$$

$$x_1^{'}-x_1^{''}+x_2^{'}-x_2^{''}+s_2=6$$

$$x_1^{'}-x_1^{''}+3x_2^{'}-3x_2^{''}+s_3=9$$

$$x_1^{'},x_2^{'},x_1^{''},x_2^{''},s_1,s_2,s_3\ge 0$$

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| Basic | $x_1^{'}$ | $x_1^{''}$ | $x_2^{'}$ | $x_2^{''}$ | $s_1$ | $s_2$ | $s_3$ | Sol | R/C |
|---|---|---|---|---|---|---|---|---|---|
| z | -2 | 2 | -3 | 3 | 0 | 0 | 0 | 0 | |
| $s_1$ | -1 | 1 | 2 | -2 | 1 | 0 | 0 | 4 | $4/2=2$ |
| $s_2$ | 1 | -1 | 1 | -1 | 0 | 1 | 0 | 6 | $6/1=6$ |
| $s_3$ | 1 | -1 | 3 | -3 | 0 | 0 | 1 | 9 | $9/3=3$ |

| Basic | $x_1^{'}$ | $x_1^{''}$ | $x_2^{'}$ | $x_2^{''}$ | $s_1$ | $s_2$ | $s_3$ | Sol | |
|---|---|---|---|---|---|---|---|---|---|
| z | -2 | 2 | -3 | 3 | 0 | 0 | 0 | 0 | |
| $x_2^{'}$ | $-\frac{1}{2}$ | $\frac{1}{2}$ | 1 | -1 | $\frac{1}{2}$ | 0 | 0 | 2 | $\frac{1}{2}x_2^{'}$ |
| $s_2$ | 1 | -1 | 1 | -1 | 0 | 1 | 0 | 6 | |
| $s_3$ | 1 | -1 | 3 | -3 | 0 | 0 | 1 | 9 | |

| Basic | $x_1^{'}$ | $x_1^{''}$ | $x_2^{'}$ | $x_2^{''}$ | $s_1$ | $s_2$ | $s_3$ | Sol | |
|---|---|---|---|---|---|---|---|---|---|
| z | $-\frac{7}{2}$ | $\frac{7}{2}$ | 0 | 0 | $\frac{3}{2}$ | 0 | 0 | 6 | $z+3x_2^{'}$ |
| $x_2^{'}$ | $-\frac{1}{2}$ | $\frac{1}{2}$ | 1 | -1 | $\frac{1}{2}$ | 0 | 0 | 2 | |
| $s_2$ | $\frac{3}{2}$ | $-\frac{3}{2}$ | 0 | 0 | $-\frac{1}{2}$ | 1 | 0 | 4 | $s_2-x_2^{'}$ |
| $s_3$ | $\frac{5}{2}$ | $-\frac{5}{2}$ | 0 | 0 | $-\frac{3}{2}$ | 0 | 1 | 3 | $s_3-x_2^{'}$ |

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| Basic | $x_1^{'}$ | $x_1^{''}$ | $x_2^{'}$ | $x_2^{''}$ | $s_1$ | $s_2$ | $s_3$ | Sol | |
|---|---|---|---|---|---|---|---|---|---|
| z | $-\frac{7}{2}$ | $\frac{7}{2}$ | 0 | 0 | $\frac{3}{2}$ | 0 | 0 | 6 | |
| $x_2^{'}$ | $-\frac{1}{2}$ | $\frac{1}{2}$ | 1 | -1 | $\frac{1}{2}$ | 0 | 0 | 2 | |
| $s_2$ | $\frac{3}{2}$ | $-\frac{3}{2}$ | 0 | 0 | $-\frac{1}{2}$ | 1 | 0 | 4 | |
| $x_1^{'}$ | 1 | -1 | 0 | 0 | $-\frac{3}{5}$ | 0 | $\frac{2}{5}$ | $\frac{6}{5}$ | $\frac{2}{5}x_1^{'}$ |

| Basic | $x_1'$ | $x_1''$ | $x_2'$ | $x_2''$ | $s_1$ | $s_2$ | $s_3$ | Sol | |
|---|---|---|---|---|---|---|---|---|---|
| z | 0 | 0 | 0 | 0 | $\frac{-3}{5}$ | 0 | $\frac{7}{5}$ | $\frac{51}{5}$ | $z+\frac{7}{2}x_1'$ |
| $x_2'$ | 0 | 0 | 1 | -1 | $\frac{1}{5}$ | 0 | $\frac{1}{5}$ | $\frac{13}{5}$ | $x_2'+\frac{1}{2}x_1'$ |
| $s_2$ | 0 | 0 | 0 | 0 | $\frac{2}{5}$ | 1 | $\frac{-3}{5}$ | $\frac{11}{5}$ | $s_2-\frac{3}{2}x_1'$ |
| $x_1'$ | 1 | -1 | 0 | 0 | $-\frac{3}{5}$ | 0 | $\frac{2}{5}$ | $\frac{6}{5}$ | |

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| Basic | $x_1'$ | $x_1''$ | $x_2'$ | $x_2''$ | $s_1$ | $s_2$ | $s_3$ | Sol | |
|---|---|---|---|---|---|---|---|---|---|
| z | 0 | 0 | 0 | 0 | $\frac{-3}{5}$ | 0 | $\frac{7}{5}$ | $\frac{51}{5}$ | |
| $x_2'$ | 0 | 0 | 1 | -1 | $\frac{1}{5}$ | 0 | $\frac{1}{5}$ | $\frac{13}{5}$ | |
| $s_1$ | 0 | 0 | 0 | 0 | 1 | $\frac{5}{2}$ | $\frac{-3}{2}$ | $\frac{11}{2}$ | $\frac{5}{2}s_1$ |
| $x_1'$ | 1 | -1 | 0 | 0 | $-\frac{3}{5}$ | 0 | $\frac{2}{5}$ | $\frac{6}{5}$ | |

| Basic | $x_1'$ | $x_1''$ | $x_2'$ | $x_2''$ | $s_1$ | $s_2$ | $s_3$ | Sol | |
|---|---|---|---|---|---|---|---|---|---|
| z | 0 | 0 | 0 | 0 | 0 | $\frac{3}{2}$ | $\frac{1}{2}$ | $\frac{27}{2}$ | $z+\frac{3}{5}s_1$ |
| $x_2'$ | 0 | 0 | 1 | -1 | 0 | $-\frac{1}{2}$ | $\frac{1}{2}$ | $\frac{3}{2}$ | |
| $s_1$ | 0 | 0 | 0 | 0 | 1 | $\frac{5}{2}$ | $\frac{-3}{2}$ | $\frac{11}{2}$ | $x_2'-\frac{1}{5}s_1$ |
| $x_1'$ | 1 | -1 | 0 | 0 | 0 | $\frac{3}{2}$ | $-\frac{1}{2}$ | $\frac{9}{2}$ | $x_1'+\frac{3}{5}s_1$ |

$$x_1^{'}=\frac{9}{2},\ x_1^{''}=0\ ,x_2^{'}=\frac{3}{2},\ x_2^{''}=0$$

As $x_1=x_1^{'}-x_1^{''}$

$$x_1=\frac{9}{2}-0$$

$$x_1=\frac{9}{2}$$

$$x_2=x_2^{'}-x_2^{''}$$

$$x_2=\frac{3}{2}-0$$

$$x_2=\frac{3}{2}$$

And z = $\frac{27}{2}$

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Lecture # 05

## Two Phase method:

In two phase method the whole procedure of solving a Linear programming problem involving artificial variable is divided into two phases.

In Phase-I we form a new objective function by assigning zero to every variable into the slack and surplus variable and negative one to each of the artificial variables (for maximization case) and multiply with positive one (for minimization case). Then we try to eliminate the artificial variables from the basis. *The solution at the phase-I serves as an initial basic feasible solution for phase-II.

In phase-II original objective function is introduced and usual simplex algorithm is used to find an optimal solution.

**Example:** Maximize $z = 2x_1 + 3x_2 + x_3$

Subject to

$$x_1 + x_2 + x_3 \le 40$$

$$2x_1 + x_2 - x_3 \ge 10$$

$$-x_2 + x_3 \ge 10$$

$$x_1, x_2, x_3 \ge 0$$

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Solution:

$$x_1 + x_2 + x_3 + s_1 = 40$$

$$2x_1 + x_2 - x_3 - s_2 = 10$$

$$-x_2 + x_3 - s_3 = 10$$

$$x_1, x_2, x_3, s_1, s_2, s_3 \ge 0$$

$$x_1 + x_2 + x_3 + s_1 = 40$$

$$2x_1 + x_2 - x_3 - s_2 + A_1 = 10$$

$$-x_2 + x_3 - s_3 + A_2 = 10$$

$$x_1, x_2, x_3, s_1, s_2, s_3, A_1, A_2 \ge 0$$

At $x_1, x_2, x_3 = 0$, $s_1 = 40$

And $-s_2 = 10 \Rightarrow s_2 = -10$

$-s_3 = 10 \Rightarrow s_3 = -10$

$s_1$ is +ve and $s_2, s_3$ are -ve

So, we introduce Artificial variable for $s_2, s_3$

In phase-I we introduce a new function 'w'

In this we write 'Artificial variable' to R.H.S with negative.

When $A_1$ or $A_2$ leave we will omit $A_1$ or $A_2$ Column.

**Phase -I:** We introduce

$$w = -A_1 - A_2$$

$$w + (A_1 + A_2) = 0 \qquad \text{_____(1)}$$

From above $A_1 = 10 - 2x_1 - x_2 + x_3 + s_2$

$$A_2 = 10 + x_2 - x_3 + s_3$$

$$A_1 + A_2 = 20 - 2x_1 + s_2 + s_3$$

Put in (1) $w + 20 - 2x_1 + s_2 + s_3 = 0$

$$w - 2x_1 + s_2 + s_3 = -20$$

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $A_1$ | $A_2$ | Sol |
|---|---|---|---|---|---|---|---|---|---|
| w | −2 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | −20 |
| $s_1$ | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 40 |
| $A_1$ | 2 | 1 | −1 | 0 | −1 | 0 | 1 | 0 | 10 |
| $A_2$ | 0 | −1 | 1 | 0 | 0 | −1 | 0 | 1 | 10 |

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| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $A_2$ | Sol |
|---|---|---|---|---|---|---|---|---|
| w | 0 | 1 | −1 | 0 | 0 | 1 | 0 | −20 |
| $s_1$ | 0 | $\frac{1}{2}$ | $\frac{3}{2}$ | 1 | $\frac{1}{2}$ | 0 | 0 | 35 |
| $x_1$ | 1 | $\frac{1}{2}$ | $-\frac{1}{2}$ | 0 | $-\frac{1}{2}$ | 0 | 0 | 5 |
| $A_2$ | 0 | −1 | 1 | 0 | 0 | −1 | 1 | 10 |

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | Sol |
|---|---|---|---|---|---|---|---|
| w | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| $s_1$ | 0 | 2 | 0 | 1 | $\frac{1}{2}$ | $\frac{3}{2}$ | 20 |
| $x_1$ | 1 | 0 | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 10 |
| $x_3$ | 0 | –1 | 1 | 0 | 0 | –1 | 10 |

In this method it is obvious that when we solve phase-I then row of 'w' will become zero. Then we start phase-II in which we put original function of 'z' remaining table will not be changed.

$$z = 2x_1 + 3x_2 + x_3$$

$$z - 2x_1 - 3x_2 - x_3 = 0$$

**Phase-II:**

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| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | Sol |
|---|---|---|---|---|---|---|---|
| z | –2 | –3 | –1 | 0 | 0 | 0 | 0 |
| $s_1$ | 0 | 2 | 0 | 1 | $\frac{1}{2}$ | $\frac{3}{2}$ | 20 |
| $x_1$ | 1 | 0 | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 10 |
| $x_3$ | 0 | –1 | 1 | 0 | 0 | –1 | 10 |

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | Sol |
|---|---|---|---|---|---|---|---|
| z | –2 | 0 | –1 | $\frac{3}{2}$ | $\frac{3}{4}$ | $\frac{9}{4}$ | 30 |
| $x_2$ | 0 | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | 10 |
| $x_1$ | 1 | 0 | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 10 |
| $x_3$ | 0 | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ | $-\frac{1}{4}$ | 20 |

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| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | Sol |
|---|---|---|---|---|---|---|---|
| z | 0 | 0 | –1 | $\frac{3}{2}$ | $-\frac{1}{4}$ | $\frac{5}{4}$ | 50 |
| $x_2$ | 0 | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | 10 |
| $x_1$ | 1 | 0 | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 10 |
| $x_3$ | 0 | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ | $-\frac{1}{4}$ | 20 |

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | Sol |
|---|---|---|---|---|---|---|---|
| z | 0 | 0 | 0 | 2 | 0 | 1 | 70 |
| $x_2$ | 0 | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | 10 |
| $x_1$ | 1 | 0 | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 10 |
| $x_3$ | 0 | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ | $-\frac{1}{4}$ | 20 |

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Here $x_1 = 10$ , $x_2 = 10$ , $x_3 = 20$ and $z = 70$ is the required solution.

Lecture # 06

## Big-M Method or M-technique:

Big-M method is a modified form of two phase method. Here M is a very very large positive value.

As M-technique is modified form of "Two phase". In Two phase method we use a new function 'w' in phase-I and original function 'z' in phase-II. But in M-technique we use new function and original function. We multiply artificial function with 'M' and subtract it from 'z'.

**Example:** Maximize $z = 2x_1 + 3x_2 + x_3$

Subject to $x_1 + x_2 + x_3 \le 40$

$$2x_1 + x_2 - x_3 \ge 10$$

$$-x_2 + x_3 \ge 10$$

$$x_1, x_2, x_3 \ge 0$$

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Solution:

$$x_1 + x_2 + x_3 + s_1 = 40$$

$$2x_1 + x_2 - x_3 - s_2 + A_1 = 10$$

$$-x_2 + x_3 - s_3 + A_2 = 10$$

$$x_1, x_2, x_3, s_1, s_2, s_3, A_1, A_2 \ge 0$$

$$z = 2x_1 + 3x_2 + x_3 - MA_1 - MA_2$$

$$z = 2x_1 + 3x_2 + x_3 - M(A_1 + A_2) \quad \text{_____(i)}$$

$$A_1 = 10 - 2x_1 - x_2 + x_3 + s_2$$

$$A_2 = 10 + x_2 - x_3 + s_3$$

$$A_1 + A_2 = 20 - 2x_1 + s_2 + s_3 \quad \text{put in (i)}$$

$$z = 2x_1 + 3x_2 + x_3 - M(20 - 2x_1 + s_2 + s_3)$$

$$z = (2 + 2M)x_1 + 3x_2 + x_3 - Ms_2 - Ms_3 - 20M$$

$$z-(2+2M)x_1-3x_2-x_3+Ms_2+Ms_3=-20M$$

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $A_1$ | $A_2$ | Sol |
|---|---|---|---|---|---|---|---|---|---|
| z | –(2+2M) | –3 | –1 | 0 | M | M | 0 | 0 | –20M |
| $s_1$ | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 40 |
| $A_1$ | 2 | 1 | –1 | 0 | –1 | 0 | 1 | 0 | 10 |
| $A_2$ | 0 | –1 | 1 | 0 | 0 | –1 | 0 | 1 | 10 |

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | $A_2$ | Sol |
|---|---|---|---|---|---|---|---|---|
| z | 0 | –2+M | –2–M | 0 | –1 | M | 0 | –10M+10 |
| $s_1$ | 0 | $\frac{1}{2}$ | $\frac{3}{2}$ | 1 | $\frac{1}{2}$ | 0 | 0 | 35 |
| $x_1$ | 1 | $\frac{1}{2}$ | $\frac{1}{2}$ | 0 | $-\frac{1}{2}$ | 0 | 0 | 5 |
| $A_2$ | 0 | –1 | 1 | 0 | 0 | –1 | 1 | 10 |

| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | Sol |
|---|---|---|---|---|---|---|---|
| z | 0 | –4 | 0 | 0 | –1 | –2 | 30 |
| $s_1$ | 0 | 2 | 0 | 1 | $\frac{1}{2}$ | $\frac{3}{2}$ | 20 |
| $x_1$ | 1 | 0 | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 10 |
| $x_3$ | 0 | –1 | 1 | 0 | 0 | –1 | 10 |

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| Basic | $x_1$ | $x_2$ | $x_3$ | $s_1$ | $s_2$ | $s_3$ | Sol |
|---|---|---|---|---|---|---|---|
| z | 0 | 0 | 0 | 2 | 0 | 1 | 70 |
| $x_2$ | 0 | 1 | 0 | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | 10 |
| $x_1$ | 1 | 0 | 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | 10 |
| $x_3$ | 0 | 0 | 1 | $\frac{1}{2}$ | $\frac{1}{4}$ | $-\frac{1}{4}$ | 20 |

$\Rightarrow \quad x_1 = 10 \ , \ x_2 = 10 \ , \ x_3 = 20 \quad$ and $\ z = 70$

## Revised Simplex method or Dual simplex method:

**Leaving variable:** Most negative value in solution column

**Entering variable:** Minimum positive ratio in Ratio-Row

This method is reciprocal of simplex method. In this method first, we choose leaving variable the greater and negative value from solution column will leave. Then we choose entering variable by making Ratio-Row. In this we choose least positive value.

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**Example:** Maximize $z = 80x_1 + 100x_2$

Subject to $80x_1 + 60x_2 \geq 1500$

$20x_1 + 90x_2 \geq 1200$

$x_1, x_2 \geq 0$

Solution: $-80x_1 - 60x_2 \leq -1500$

$-20x_1 - 90x_2 \leq -1200$

$x_1, x_2 \geq 0$

$-80x_1 - 60x_2 + s_1 = -1500$

$-20x_1 - 90x_2 + s_2 = -1200$

$x_1, x_2, s_1, s_2 \geq 0$ and $z - 80x_1 - 100x_2 - 0s_1 - 0s_2 = 0$

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol |
|---|---|---|---|---|---|
| z | $-80$ | $-100$ | 0 | 0 | 0 |
| $s_1$ | $-80$ | $-60$ | 1 | 0 | $-1500$ |
| $s_2$ | $-20$ | $-90$ | 0 | 1 | $-1200$ |
| $R/R$ | $-80/80 = 1$ | $\frac{-100}{-60} = 5/3$ | 0 | 0 | |

$\because R/R = \frac{z}{leaving\ row}$

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol |
|---|---|---|---|---|---|
| z | 0 | $-40$ | $-1$ | 0 | 1500 |
| $x_1$ | 1 | $\frac{3}{4}$ | $-\frac{1}{80}$ | 0 | $\frac{75}{4}$ |
| $s_2$ | 0 | $-75$ | $-\frac{1}{4}$ | 1 | $-825$ |
| $R/R$ | 0 | $\frac{-40}{-75} = 8/15$ | $\frac{-1}{-\frac{1}{4}} = 4$ | 0 | |

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| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | Sol |
|---|---|---|---|---|---|
| z | 0 | 0 | $-\frac{13}{15}$ | $\frac{-8}{15}$ | 1940 |
| $x_1$ | 1 | 0 | $-\frac{3}{200}$ | $\frac{1}{100}$ | $\frac{21}{2}$ |
| $x_2$ | 0 | 1 | $\frac{1}{300}$ | $-\frac{1}{75}$ | 11 |

$\Rightarrow \quad x_1 = \frac{21}{2}$ , $x_2 = 11$ and $z = 1940$

**Note**: Initial solution does not exist in Revised simplex method.

Lecture # 07

## Duality in Linear Programming:

Duality is very important concept associated with linear programming. The term duality implies that every linear programming problem whether of maximize or minimize is associated with another linear programming problem based on the same data.

The original problem is this context is called primal problem whereas the other is called its dual problem.

**Example:** Maximize $z = x_1 + x_2 + x_3$

Subject to $x_1 - 2x_2 + x_3 \geq 4$

$2x_1 + x_2 - 3x_3 \geq 3$

$-x_1 + x_2 - x_3 \geq 4$

$x_1, x_2, x_3 \geq 0$

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Its dual problem

$z = 4y_1 + 3y_2 + 4y_3$

Subject to $y_1 + 2y_2 - y_3 \leq 1$

$-2y_1 + y_2 + y_3 \leq 1$

$y_1 - 3y_2 - y_3 \leq 2$

$y_1, y_2, y_3 \geq 0$

For MCQ *Dual of dual problem is called primal or oringnal problem.

## Transportation problem:

The transportation problem deals with a special class of linear programming problem in which the objective is to transport a product manufactured at several plants to a number of different destination at a minimum total cost. The quantity demanded by the destination are given in the statement of the problem. The cost of shipping a unit of goods from a non-origin to a non-destination is also given. Our objective is to detemive the total minimum shipping cost.

## Basic feasible solution:

A solution of M × N transportation problem is said to be basic feasible solution if the total number of allocation is equal to m + n – 1

## Optimal Solution:

A basic feasible solution is said to be optimal when the total transportaion cost is minimum.

## Methods for finding basic feasible solution:

There are three methods for finding basic feasible solution

(i) North-West corner method (NWCM) or Top left corner mehtod.
(ii) Least cost method (LCM) or Matrix minimum method.
(iii) Vogel's approximation mehtod (VAM) or Method of penalty

### (i) North-West corner method:

### Procedure of NWCM:

(i) First we check the problem is balance or unbalance. If supply equal demands then the problem is balance otherwise unbalance.
(ii) Select the topleft corner cell (box) of the transportaion problem and allocates as many unit as possible equal to the minimum between available supply and demand.
(iii) Adjust the supply and demand number in the respective rows and column.
(iv) If the demand for the first cell is satisfied then move horizontally to the next cell in the second column.
(v) If the supply for the first cell is satisfied then move down in the second row.
(vi) If demand is satisfied then delete (×) the column.
(vii) If supply is satisfied then delete (×) the row.
(viii) Continue the process until all supply and demand are satisfied.

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**Example:**

| Factory | Distribution | | | | |
|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | Supply |
| 1 | $200\boxed{3}$ | $50\boxed{1}$ | $\times\boxed{7}$ | $\times\boxed{9}$ | ~~250~~ ~~50~~ × |
| 2 | $\times\boxed{2}$ | $250\boxed{6}$ | $100\boxed{5}$ | $\times\boxed{9}$ | ~~350~~ ~~100~~ × |
| 3 | $\times\boxed{8}$ | $\times\boxed{3}$ | $250\boxed{3}$ | $150\boxed{2}$ | ~~400~~ ~~150~~ × |
| Demand | ~~200~~ × | ~~300~~ ~~250~~ × | ~~350~~ ~~250~~ × | ~~150~~ × | 1000 |

Total no. of allocations = m + n – 1

$6 = 3+4 - 1$

$6 = 6$

Total cost = 200 × 3 + 50 ×1 + 250×6+ 100×5+250×3+150×2 = 3700

**Question:**

| | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|---|---|---|---|---|---|
| $O_1$ | $60\boxed{20}$ | $40\boxed{22}$ | $20\boxed{17}$ | $\times\boxed{4}$ | ~~120~~ ~~60~~ ~~20~~ × |
| $O_2$ | $\times\boxed{24}$ | $\times\boxed{37}$ | $10\boxed{9}$ | $60\boxed{7}$ | ~~70~~ ~~60~~ × |
| $O_3$ | $\times\boxed{32}$ | $\times\boxed{37}$ | $\times\boxed{20}$ | $50\boxed{15}$ | ~~50~~ × |
| Demand | ~~60~~ × | ~~40~~× | ~~30~~ ~~10~~ × | ~~110~~ ~~50~~× | 240 |

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Total no. of allocations = m + n – 1

$6 = 3 + 4 - 1$

$6 = 6$

Total cost = 60×20 + 40×22 + 20×17 + 10×9 +60×7+50×15 = 3680

**Question:**

| | $w_1$ | $w_2$ | $w_3$ | Supply |
|---|---|---|---|---|
| $P_1$ | $10\boxed{2}$ | $\times\boxed{1}$ | $\times\boxed{5}$ | ~~10~~ × |
| $P_2$ | $5\boxed{7}$ | $20\boxed{3}$ | $\times\boxed{7}$ | ~~25~~ ~~20~~ × |
| $P_3$ | $\times\boxed{6}$ | $2\boxed{5}$ | $18\boxed{15}$ | ~~20~~ ~~18~~ × |
| Demand | ~~15~~ ~~5~~ × | ~~22~~ ~~2~~ × | ~~18~~ × | 55 |

Total no. of allocations = m + n – 1 $\Rightarrow$ 5 = 3 + 3 – 1 $\Rightarrow$ 5 = 5

Total cost = 10×2 + 5×7 + 20×3 + 2×5 +18×3 = 179

**Question:** Construct 4×5 transportation problem (balanced) and find basic feasible solution by Top-left corner method.

| | $w_1$ | $w_2$ | $w_3$ | $w_4$ | $w_5$ | Supply |
|---|---|---|---|---|---|---|
| $P_1$ | $55\boxed{2}$ | $22\boxed{5}$ | $23\boxed{4}$ | $\times\boxed{7}$ | $\times\boxed{3}$ | ~~100 45 23~~ × |
| $P_2$ | $\times\boxed{1}$ | $\times\boxed{3}$ | $22\boxed{4}$ | $3\boxed{2}$ | $\times\boxed{6}$ | ~~25 3~~ × |
| $P_3$ | $\times\boxed{4}$ | $\times\boxed{2}$ | $\times\boxed{5}$ | $25\boxed{3}$ | $15\boxed{2}$ | ~~40 15~~ × |
| $P_4$ | $\times\boxed{3}$ | $\times\boxed{5}$ | $\times\boxed{4}$ | $\times\boxed{1}$ | $35\boxed{2}$ | ~~35~~ × |
| Demand | ~~55~~ × | ~~22~~ × | ~~45 22~~× | ~~28~~ 25 × | ~~50 35~~ × | 200 |

Total no. of allocations = m + n – 1

8 = 4 + 5 – 1

8 = 8

Total cost = 55×2 + 22×5 + 23×4 + 22×4 +3×2 + 25×3+15×2+35×2

= 581

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Lecture # 08

## Special Case Degenerate / Degeneracy:

Degeneracy occurs when (Total no. of allocations ≠ m + n – 1)

## Question:

| | X | Y | Z | Supply |
|---|---|---|---|---|
| A | $\lceil 8 \rceil$ | $\lceil 7 \rceil$ | $\lceil 3 \rceil$ | 60 |
| B | $\lceil 3 \rceil$ | $\lceil 8 \rceil$ | $\lceil 9 \rceil$ | 70 |
| C | $\lceil 11 \rceil$ | $\lceil 3 \rceil$ | $\lceil 5 \rceil$ | 80 |
| Demand | 50 | 80 | 80 | 210 |

Solution:

| | X | Y | Z | Supply |
|---|---|---|---|---|
| A | 50 $\lceil 8 \rceil$ | 10 $\lceil 7 \rceil$ | × $\lceil 3 \rceil$ | ~~60~~ ~~10~~ × |
| B | × $\lceil 3 \rceil$ | 70 $\lceil 8 \rceil$ | × $\lceil 9 \rceil$ | ~~70~~ × |
| C | × $\lceil 11 \rceil$ | d $\lceil 3 \rceil$ | 80 $\lceil 5 \rceil$ | ~~80~~ × |
| Demand | ~~50~~ × | ~~80~~ ~~70~~ × | ~~80~~ × | 210 |

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Since d is very very small positive number whose contribution in the solution is negligible.

Total no. of allocations = m + n – 1

5 = 3+3 – 1

5 = 5

Total cost = 50 × 8 + 10 ×7 + 70×8+3d + 80×5 = 1430+3d

## Unbalance problem:

| | 5 | 6 | 7 | 8 | 9 | 10 | Supply |
|---|---|---|---|---|---|---|---|
| 1 | $\lceil 9 \rceil$ | $\lceil 12 \rceil$ | $\lceil 9 \rceil$ | $\lceil 6 \rceil$ | $\lceil 9 \rceil$ | $\lceil 10 \rceil$ | 5 |
| 2 | $\lceil 7 \rceil$ | $\lceil 3 \rceil$ | $\lceil 7 \rceil$ | $\lceil 7 \rceil$ | $\lceil 5 \rceil$ | $\lceil 5 \rceil$ | 6 |
| 3 | $\lceil 6 \rceil$ | $\lceil 5 \rceil$ | $\lceil 9 \rceil$ | $\lceil 11 \rceil$ | $\lceil 3 \rceil$ | $\lceil 11 \rceil$ | 2 |
| 4 | $\lceil 6 \rceil$ | $\lceil 8 \rceil$ | $\lceil 11 \rceil$ | $\lceil 2 \rceil$ | $\lceil 2 \rceil$ | $\lceil 10 \rceil$ | 2 |
| Demand | 4 | 4 | 6 | 2 | 4 | 2 | |

- If supply less than demand then we add row
- If demand less than supply then we add column

Solution:

| | 5 | 6 | 7 | 8 | 9 | 10 | Supply |
|---|---|---|---|---|---|---|---|
| 1 | 4 \|9 | 1 \|12 | × \|9 | × \|6 | × \|9 | × \|10 | ~~5~~ × |
| 2 | × \|7 | 3 \|3 | 3\|7 | × \|7 | × \|5 | × \|5 | ~~6~~ ~~3~~ × |
| 3 | × \|6 | × \|5 | 2\|9 | × \|11 | × \|3 | × \|11 | ~~2~~ × |
| 4 | × \|6 | × \|8 | 1\|11 | 1\|2 | × \|2 | × \|10 | ~~2~~ ~~1~~ × |
| Unsatisfied Row | × \|0 | × \|0 | × \|0 | 1\|0 | 4\|0 | 2\|0 | ~~7~~ ~~6~~ ~~2~~ × |
| Demand | ~~4~~ × | ~~4~~ ~~3~~ × | ~~6~~ ~~3~~ ~~1~~ × | ~~2~~ ~~1~~ × | ~~4~~ × | ~~2~~ × | 22 |

No. of allocations = m + n – 1

10 = 5 + 6 –1

10 = 10

Total cost = 4×9 + 1×12 + 3×3 + 3×7 + 2×9 + 1×11 + 1×2 + 1× 0+4×0+2×0

= 109

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## Least Cost Method:

## Procedure:

(i) First of all we check the problem is balance or unbalance.

(ii) Identify the box having minimum unit transportation cost $(c_{ij})$

(iii) If the minimum cost is not unique then choose the top left corner box for allocation.

(iv) Choose the value of the corresponding $x_{ij}$ as much as possible subject to the supply and demand constraints.

(v) Repeat the above steps untill all restrictions are satisfied.

**Example:**

| | 5 | 6 | 7 | 8 | 9 | 10 | Supply |
|---|---|---|---|---|---|---|---|
| 1 | × \|9 | × \|12 | 5 \|9 | × \|6 | × \|9 | × \|10 | ~~5~~ × |
| 2 | × \|7 | 1 \|3 | 1\|7 | × \|7 | 2 \|5 | 2 \|5 | ~~6 5 3 1~~ × |
| 3 | × \|6 | × \|5 | ×\|9 | × \|11 | 2 \|3 | × \|11 | ~~2~~ × |
| 4 | × \|6 | × \|8 | ×\|11 | 2\|2 | d \|2 | × \|10 | ~~2~~ × |
| Unsatisfied Row | 4 \|0 | 3 \|0 | × \|0 | ×\|0 | ×\|0 | ×\|0 | ~~7 3~~ × |
| Demand | ~~4~~ × | ~~4 1~~ × | ~~6 5~~ × | ~~2~~ × | ~~4 2~~ × | ~~2~~ × | 22 |

Number of allocations = m + n – 1

10 = 5 + 6 – 1

Total cost = 5×9 + 1×3 + 1×7 + 2×5 + 2×5 + 2×3 + 2×2 + 2d + 4×0 + 3×0

= 85 + 2d

**Question:**

Find basic feasible solution of the following transportaion problem by Least cost method.

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| | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| 1 | ×\|3 | 250\|1 | ×\|7 | ×\|4 | ~~250~~ × |
| 2 | 200\|2 | ×\|6 | 150\|5 | ×\|9 | ~~350 150~~ × |
| 3 | ×\|8 | 50\|3 | 200\|3 | 150\|2 | ~~400 250 200~~ × |
| Demand | ~~200~~ × | ~~300 50~~ × | ~~350 150~~ × | ~~150~~ × | 1000 |

Total no. of allocations = m + n – 1

6 = 3 + 4 – 1

6 = 6

Total cost = 250×1 + 200×2 + 150×5 + 50×3 +200×3+150×2 = 2450

**Question:**

Find basic feasible solution of the following transportaion problem by Least cost method.

| | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|---|---|---|---|---|---|
| $O_1$ | 10 \|20 | ×\|22 | ×\|17 | 110\|4 | ~~120~~ ~~10~~ × |
| $O_2$ | 40 \|24 | ×\|37 | 30\|9 | ×\|7 | ~~70~~ ~~40~~ × |
| $O_3$ | 10 \|32 | 40\|37 | ×\|20 | ×\|15 | ~~50~~ ~~40~~ × |
| Demand | ~~60~~ ~~50~~ ~~10~~ × | ~~40~~× | 30 ~~×~~ | ~~110~~ × | 240 |

Total no. of allocations = m + n – 1

6 = 3 + 4 – 1

6 = 6

Total cost = 10×20 + 110×4 + 40×24 + 30×9 +10×32+40×37 = 7630

**Question:**

Find basic feasible solution of the following transportaion problem by Least cost method.

| | $w_1$ | $w_2$ | $w_3$ | Supply |
|---|---|---|---|---|
| $P_1$ | ×\|2 | 10\|1 | ×\|5 | ~~10~~ × |
| $P_2$ | 13\|7 | 12\|3 | ×\|4 | ~~25~~ ~~13~~ × |
| $P_3$ | 2\|6 | ×\|5 | 18\|3 | ~~20~~ ~~2~~ × |
| Demand | ~~15~~ ~~13~~× | ~~22~~ ~~12~~ × | ~~18~~ × | 55 |

Total no. of allocations = m + n – 1

5 = 3 + 3 – 1

5 = 5

Total cost = 10×1 + 13×7 + 12×3 + 2×6 +18×3 = 203

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**Question:**

Find basic feasible solution of the following transportaion problem by Least cost method.

| | $w_1$ | $w_2$ | $w_3$ | $w_4$ | $w_5$ | Supply |
|---|---|---|---|---|---|---|
| $P_1$ | $30\boxed{2}$ | $\times\boxed{5}$ | $45\boxed{4}$ | $\times\boxed{7}$ | $25\boxed{3}$ | ~~100 70 45~~ × |
| $P_2$ | $25\boxed{1}$ | $\times\boxed{3}$ | $\times\boxed{4}$ | $\times\boxed{2}$ | $\times\boxed{6}$ | ~~25~~ × |
| $P_3$ | $\times\boxed{4}$ | $22\boxed{2}$ | $\times\boxed{5}$ | $\times\boxed{3}$ | $18\boxed{2}$ | ~~40 18~~ × |
| $P_4$ | $\times\boxed{3}$ | $\times\boxed{5}$ | $\times\boxed{4}$ | $28\boxed{1}$ | $7\boxed{2}$ | ~~35 7~~ × |
| Demand | ~~55 30~~ × | ~~22~~ × | ~~45~~ × | ~~28~~ × | ~~50 32 25~~× | 200 |

Total no. of allocations = m + n – 1

$8 = 4 + 5 - 1$

$8 = 8$

Total cost = $30\times2 + 45\times4 + 25\times3 + 25\times1 + 22\times2 + 18\times2 + 28\times1 + 7\times2$

$= 462$

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Lecture # 09

## Vogel's Approximation method or Method of Penalty:

The Vogel's approximation method is an iterative procedure for computing an initial basic feasible solution of a transportation problem. This method is preferred over the two methods discussed in the previous sections, because the initial basis feasible solution obtained by this method is either optimal or very close to optimal solution.

"This method is a little complex than the previously discussed methods. So, go slowly and reread the explanation at least twice".

## Steps in Vogel's Approximation method (VAM):

(i) First of all, we check our problem is balance or unbalance.

(ii) Identify the boxes having minimum and next to minimum transportation cost in each row and write the difference (penalty) along the side of the table against the corresponding row.

(iii) Identify the boxes having minimum and next to minimum transportation cost in each column and write the difference (penalty) against the corresponding column.

(iv) Identify the maximum penalty. If it is along the side of the table, make maximum allocation to the box having minimum cost of transportation in that row. If it is below the table, make maximum allocation to the box having minimum cost of transportation in that column.

(v) If the penalties corresponding to two or more rows or columns are equal you are at liberty to break the tie arbitrarily.

(vi) Repeat the above steps until all restrictions are satisfied.

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**Question:** Find basic feasible solution by Vogel's approximation method of the following transportation problem.

| | 4 | 5 | 6 | 7 | Supply |
|---|---|---|---|---|---|
| 1 | × 4 | 6 3 | 18 0 | × 5 | ~~24~~ 6 × |
| 2 | × 1 | 13 2 | × 6 | 4 1 | ~~17~~ 13 × |
| 3 | 15 3 | × 6 | × 2 | 4 3 | ~~19~~ 4 × |
| Demand | ~~15~~ × | ~~19~~ ~~13~~ × | ~~18~~ × | ~~8~~ ~~4~~ × | |

| | | | | |
|---|---|---|---|---|
| (3) | 1 | 2 | 2 | (3) |
| 1 | 1 | 1 | 1 | 2 |
| 1 | (3) | (3) | - | - |

| | | | |
|---|---|---|---|
| 2 | 1 | 2 | 2 |
| 2 | 1 | - | 2 |
| - | 1 | - | 2 |
| - | 1 | - | (4) |
| - | 1 | - | - |

Total No. of allocations = m + n – 1

6 = 3 + 4 – 1

6 = 6

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Total cost = 6 × 3 + 18 × 0 +13×2 + 4×1 + 15×3 + 4×3

= 105

**Question:** Find basic feasible solution by North West corner method, Least corner method, Vogel's approximation method of the following transportation problem. **Solution**: (i) By NWCM

| | X | Y | Z | Supply |
|---|---|---|---|---|
| A | 50 8 | 10 7 | × 3 | ~~60~~ ~~10~~ × |
| B | × 3 | 70 8 | × 9 | ~~70~~ × |
| C | × 11 | d 3 | 80 5 | ~~80~~ × |
| Demand | ~~50~~ × | ~~80~~ ~~70~~ × | ~~80~~ × | 210 |

Since d is very very small positive number whose contribution in the solution is negligible. Total no. of allocations = m + n – 1

5 = 3+3 – 1

5 = 5

Total cost = 50 × 8 + 10 ×7 + 70×8+3d + 80×5 = 1430+3d

(ii) By L.C.M

| | X | Y | Z | Supply |
|---|---|---|---|---|
| A | × 8 | ×7 | 60 3 | ~~60~~ × |
| B | 50 3 | ×8 | 20 9 | 70 ~~20~~ × |
| C | × 11 | 80 3 | d 5 | ~~80~~ × |
| Demand | ~~50~~ × | ~~80~~ × | ~~80~~ 20 × | 210 |

Total no. of allocations = m + n – 1

5 = 3+3 – 1

5 = 5

Total cost = 60 × 3 + 50 ×3 + 20 × 9 + 80 × 3 + 5d = 750 + 5d

(iii) By Vogel's Approximation method:

| | X | Y | Z | Supply |
|---|---|---|---|---|
| A | × 8 | ×7 | 60 3 | ~~60~~ × |
| B | 50 3 | ×8 | 20 9 | 70 ~~20~~ × |
| C | × 11 | 80 3 | d 5 | ~~80~~ × |
| Demand | ~~50~~ × | ~~80~~ × | ~~80~~ 20 × | 210 |

| | | | |
|---|---|---|---|
| 4 | (4) | - | |
| (5) | 1 | 1 | |
| 2 | 2 | 2 | |

5 4 2

- 4 2

- (5) 4

Total No. of allocations = m + n – 1

5 = 3 + 3 – 1

5 = 5

Total cost = 60 × 3 + 50 × 3 + 20×9 + 80 × 3 + 5d

= 750 + 5d

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**Question:** Find basic feasible solution by Vogel's approximation method of the following transportation problem.

| | $w_1$ | $w_2$ | $w_3$ | Supply |
|---|---|---|---|---|
| A | $\boxed{28}$ | $\boxed{17}$ | $\boxed{26}$ | 500 |
| B | $\boxed{19}$ | $\boxed{12}$ | $\boxed{16}$ | 300 |
| Demand | 250 | 250 | 500 | |

Solution:

| | $w_1$ | $w_2$ | $w_3$ | Supply |
|---|---|---|---|---|
| A | 50 $\boxed{28}$ | 250 $\boxed{17}$ | 200 $\boxed{26}$ | ~~500~~ ~~450~~ ~~250~~ × |
| B | × $\boxed{19}$ | × $\boxed{12}$ | 300 $\boxed{16}$ | ~~300~~ × |
| Unrestricted row | 200 $\boxed{0}$ | × $\boxed{0}$ | × $\boxed{0}$ | ~~200~~ × |
| Demand | ~~250~~ ~~50~~ × | ~~250~~ × | ~~500~~ ~~200~~ × | |

| | | | |
|---|---|---|---|
| 9 | 9 | 9 | 9 |
| 4 | 4 | - | - |
| 0 | - | - | - |

(19) 12 16

9 5 (10)

(28) 17 26

\- 17 (26)

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Total No. of allocations = m + n – 1

5 = 3 + 3 – 1

5 = 5

Total cost = 50 × 28 + 250 × 17 + 200×26 + 300 × 16 + 200 × 0

= 15650

Lecture # 10

## Optimal Solution for transportation problem:

There are two methods for finding an optimal solution. For transportation problem

(i) Stepping stone method
(ii) MODI method or Modified Distribution method or u-v method

### (i) Stepping Stone method:

**Question:** Find initial basic feasible solution by Least Cost Method (L.C.M) and optimal solution by Stepping stone method of the following transportation problem.

Solution:

| | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| A | × 4 | 450 6 | × 8 | 250 6 | ~~700~~ ~~250~~ × |
| B | 50 3 | × 5 | 350 2 | × 5 | ~~400~~ ~~50~~ × |
| C | 350 3 | × 9 | × 6 | 250 5 | ~~600~~ ~~250~~ × |
| Demand | ~~400~~ ~~350~~ × | ~~450~~ × | ~~350~~ × | ~~500~~ ~~250~~ × | 1700 |

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Total cost = 450 × 6 + 250 × 6 + 50×3 + 350 × 2 + 350 × 3 + 250 × 5

= 7350

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 11 | 4 – 6 + 5 – 3 = 0 | NIND |
| 13 | 8 – 6 + 5 – 3 + 3 – 2 = 5 | Increase |
| 22 | 5 – 6 + 6 – 5 + 3 – 3 = 0 | NIND |
| 24 | 5 – 5 + 3 – 3 = 0 | NIND |
| 32 | 9 – 6 + 6 – 5 = 4 | Increase |
| 33 | 6 – 2 + 3 – 3 = 4 | Increase |

Since all the values of unoccupied cells are non-negative. So, the solution obtained is optimal.

**Question:** Find initial basic feasible solution by Least Cost Method (L.C.M) and optimal solution by Stepping stone method of the following transportation problem.

Solution:

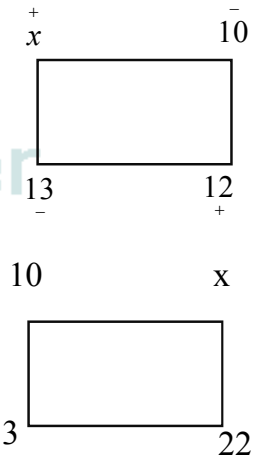
| | 1 | 2 | 3 | Supply |
|---|---|---|---|---|
| A | × [2] | 10 [1] | × [5] | 10 × |
| B | 13 [7] | 12 [3] | × [4] | ~~13~~ ~~25~~ × |
| C | 2 [6] | × [5] | 18 [3] | ~~20~~ ~~2~~ × |
| Demand | ~~15~~ ~~13~~ × | ~~22~~ ~~12~~ × | ~~18~~ × | |

Total cost = 10 × 1 + 13 × 7 + 12×3 + 2 × 6 + 18 × 3 = 203

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 11 | 2 – 1 + 3 – 7 = –3 | Decrease |
| 13 | 5 – 3 + 6 – 7 + 3 – 1 = 3 | Increase |
| 23 | 4 – 3 + 6 – 7 = 0 | NIND |
| 32 | 5 – 6 + 7 – 3 = 3 | Increase |

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| | 1 | 2 | 3 | Supply |
|---|---|---|---|---|
| A | 10 [2] | × [1] | × [5] | 10 × |
| B | 3 [7] | 22 [3] | × [4] | ~~13~~ ~~25~~ × |
| C | 2 [6] | × [5] | 18 [3] | ~~20~~ ~~2~~ × |
| Demand | ~~15~~ ~~13~~ × | ~~22~~ ~~12~~ × | ~~18~~ × | |




Total cost = 10 × 2 + 3 × 7 + 22×3 + 2 × 6 + 18 × 3

= 173

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 12 | 1 – 3 + 7 – 2 = 3 | Increase |
| 13 | 5 – 3 + 6 – 2 = 6 | Increase |
| 23 | 4 – 3 + 6 – 7 = 0 | NIND |
| 32 | 5 – 3 + 7 – 6 = 3 | Increase |

نئے ٹیبل میں ہم جو xاس کی جگہ اور مثبت ویلیو میں چھوٹی قیمت جمع کرنا ہے۔اور منفی ویلیو کی جگہ والی رقم سے تفریق کرنا ہے۔

Since all the values of unoccupied cells are non-negative. So, sol is optimal.

**Question:** Find initial basic feasible solution by Least Cost Method (L.C.M) and optimal solution by Stepping stone method of the following transportation problem.

Solution:

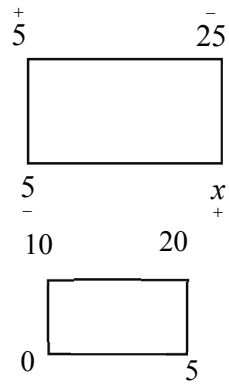
| | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| A | × 6 | 5 8 | × 8 | 25 5 | ~~30 5~~ × |
| B | 35 5 | 5 11 | × 9 | × 7 | ~~40 5~~ × |
| C | × 8 | 18 9 | 32 7 | × 13 | ~~50 18~~ × |
| Demand | ~~35~~ × | ~~28 23 18~~ × | ~~32~~ × | ~~25~~ × | 120 |

Total cost = $5 \times 8 + 25 \times 5 + 35 \times 5 + 5 \times 11 + 18 \times 9 + 32 \times 7 = 781$

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 11 | $6 - 8 + 11 - 5 = 4$ | Increase |
| 13 | $8 - 7 + 9 - 8 = 2$ | Increase |
| 23 | $9 - 7 + 9 - 11 = 0$ | NIND |
| 24 | $7 - 5 + 8 - 11 = -1$ | Decrease |
| 31 | $8 - 9 + 11 - 5 = 5$ | Increase |
| 34 | $13 - 5 + 8 - 9 = 7$ | Increase |

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| | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| A | × 6 | 10 8 | × 8 | 20 5 | ~~30 5~~ × |
| B | 35 5 | 0 11 | × 9 | 5 7 | ~~40 5~~ × |
| C | × 8 | 18 9 | 32 7 | × 13 | ~~50 18~~ × |
| Demand | ~~35~~ × | ~~28 23 18~~ × | ~~32~~ × | ~~25~~ × | 120 |




Total cost = $10 \times 8 + 20 \times 5 + 35 \times 5 + 0 \times 11 + 5 \times 7 + 18 \times 9 + 32 \times 7 = 776$

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 11 | $6 - 5 + 7 - 5 = 3$ | Increase |
| 13 | $8 - 7 + 9 - 8 = 2$ | Increase |
| 23 | $9 - 7 + 9 - 11 = 0$ | NIND |
| 31 | $8 - 9 + 11 - 5 = 5$ | Increase |
| 34 | $13 - 5 + 8 - 9 = 7$ | Increase |

Since all the values of unoccupied cells are non-negative. So, the solution obtained is optimal.

Lecture # 11

**Question:** Find basic feasible solution by matrix minimum method and optimal solution by Stepping stone method of the following transportation problem.

Solution:

| | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| A | × 3 | 250 1 | × 7 | × 4 | ~~250~~ × |
| B | 200 2 | × 6 | 150 5 | × 9 | ~~350~~ ~~150~~ × |
| C | × 8 | 50 3 | 200 3 | 150 2 | ~~400~~ ~~250~~ ~~200~~ × |
| Demand | ~~200~~ × | ~~300~~ ~~50~~ × | ~~350~~ ~~150~~ × | ~~150~~ × | 1000 |

Total cost = 250 × 1 + 200 × 2 + 150×5 + 50 × 3 + 200 × 3 + 150 × 2

= 2450

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| A1 | 3 – 1 + 3 – 3+5–2 = 5 | Increase |
| A3 | 7 – 3 + 3 – 1 = 6 | Increase |
| A4 | 4 – 2 + 3 – 1 = 4 | Increase |
| B2 | 6 – 5 + 3 – 3 = 1 | Increase |
| B4 | 9 – 2 + 3 – 5 = 5 | Increase |
| C1 | 8– 3 + 5 – 2 = 8 | Increase |

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Since all the values of unoccupied cells are non-negative. So, the solution obtained is optimal.

**Question:** Find initial basic feasible solution by Least Cost Method (L.C.M) and optimal solution by Stepping stone method of the following transportation problem.

Solution:

| | 1 | 2 | 3 | Supply |
|---|---|---|---|---|
| 1 | × 5 | × 4 | 100 3 | 100 × |
| 2 | × 8 | 200 4 | 100 3 | ~~300~~ ~~200~~ × |
| 3 | 300 9 | d 7 | × 5 | ~~300~~ × |
| Demand | ~~300~~ × | ~~200~~ × | ~~200~~ ~~100~~ × | |

Total cost = 10 × 1 + 13 × 7 + 12×3 + 2 × 6 + 18 × 3 = 4100 + 7d

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 11 | $5 - 3 + 3 - 4 + 7 - 9 = -1$ | Decrease |
| 12 | $4 - 3 + 3 - 4 = 0$ | NIND |
| 21 | $8 - 4 + 7 - 9 = 2$ | Increase |
| 33 | $5 - 3 + 4 - 7 = -1$ | Decrease |

| | 1 | 2 | 3 | Supply |
|---|---|---|---|---|
| 1 | 100 $\boxed{5}$ | $\times$ $\boxed{4}$ | $\times$ $\boxed{3}$ | 100 |
| 2 | $\times$ $\boxed{8}$ | 100 $\boxed{4}$ | 200 $\boxed{3}$ | 300 |
| 3 | 200 $\boxed{9}$ | 100+d $\boxed{7}$ | $\times$ $\boxed{5}$ | 300 |
| Demand | 300 | 200 | 200 | 700 |

Total cost = $100 \times 5 + 100 \times 4 + 200 \times 3 + 200 \times 9 + (100 + d) \times 7$

$= 4000 + 7d$

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 12 | $4 - 7 + 9 - 5 = 1$ | Increase |
| 13 | $3 - 3 + 4 - 7 + 9 - 5 = 1$ | Increase |
| 21 | $8 - 4 + 7 - 9 = 2$ | Increase |
| 33 | $5 - 3 + 4 - 7 = -1$ | Decrease |

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| | 1 | 2 | 3 | Supply |
|---|---|---|---|---|
| 1 | 100 $\boxed{5}$ | $\times$ $\boxed{4}$ | $\times$ $\boxed{3}$ | 100 |
| 2 | $\times$ $\boxed{8}$ | 200+d $\boxed{4}$ | 100–d $\boxed{3}$ | 300 |
| 3 | 200 $\boxed{9}$ | $\times$ $\boxed{7}$ | 100+d $\boxed{5}$ | 300 |
| Demand | 300 | 200 | 200 | 700 |

Total cost = $100 \times 5 + (200+d) \times 4 + (100-d) \times 3 + 200 \times 9 + (100 + d) \times 5$

$= 3900 + 6d$

| Unoccupied Cell | Increase in cost of per unit reallocation | Remarks |
|---|---|---|
| 12 | $4 - 4 + 3 - 5 + 9 - 5 = 2$ | Increase |
| 13 | $3 - 5 + 9 - 5 = 2$ | Increase |
| 21 | $8 - 3 + 5 - 9 = 1$ | Increase |
| 32 | $7 - 5 + 3 - 4 = 1$ | Increase |

Since all the values of unoccupied cells are non-negative. So, the solution is optimal.

## Modified Distribution Method or MODI Method or u-v method:

| $u_i$ | | $v_1$ 1 | $v_2$ 2 | $v_3$ 3 | Supply | |
|---|---|---|---|---|---|---|
| $u_1$ | A | × 5 | × 4 | 100 3 | 100 × | 0 |
| $u_2$ | B | × 8 | 200 4 | 100 3 | ~~300 200~~ × | 0 |
| $u_3$ | C | 300 9 | d 7 | × 5 | 300 × | 3 |
| | Demand | 300 × | 200 × | ~~200 100~~ × | 700 | |
| | | 6 | 4 | 3 | | |

Total cost = 100 × 3 + 200 × 4 + 100 × 3 + 300 × 9 + 7d = 4100 + 7d

Computing $u_i$ *and* $v_j$ by using formula

$$u_i + v_j = c_{ij} \text{ (occupied cells)}$$

In every question we take $u_1 = 0$ and find other values with help of it.

Initially we take $u_1 = 0$

$$u_1 + v_3 = 3$$

$$\because\ u_1 = 0 \qquad 0 + v_3 = 3 \ \Rightarrow v_3 = 3$$

$$u_2 + v_3 = 3$$

$$u_2 + 3 = 3 \ \Rightarrow u_2 = 0$$

$$u_2 + v_2 = 4$$

$$0 + v_2 = 4 \ \Rightarrow v_2 = 4$$

$$u_3 + v_2 = 7$$

$$u_3 + 4 = 7 \ \Rightarrow u_3 = 3$$

$$u_3 + v_1 = 9$$

$$3 + v_1 = 9 \ \Rightarrow v_1 = 6$$

Find opportunity cost by using formula $c_{ij} - \left(u_i + v_j\right)$ (unoccupied cells)

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| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| 11 | $5 - (0 + 6) = -1$ | Decrease |
| 12 | $4 - (0 + 4) = 0$ | N.I.N.D |
| 21 | $8 - (0+6) = 2$ | Increase |
| 33 | $5 - (3 + 3) = -1$ | Decrease |

| $v_j$ / $u_i$ | | 1 | 2 | 3 | Supply | |
|---|---|---|---|---|---|---|
| $u_1$ | 1 | 100 [5] | × [4] | × [3] | 100 | 0 |
| $u_2$ | 2 | × [8] | 100 [4] | 200 [3] | 300 | 1 |
| $u_3$ | 3 | 200 [9] | 100+d [7] | × [5] | 300 | 4 |
| | Demand | 300 | 200 | 200 | 700 | |
| | | 5 | 3 | 2 | | |

Total cost = $100 \times 5 + 100 \times 4 + 200 \times 3 + 200 \times (100+d) \times 7 = 4000 + 7d$

Initially we take $u_1 = 0$

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$$u_1 + v_1 = 5$$

$$\because\ u_1 = 0 \qquad 0 + v_1 = 5 \ \Rightarrow v_1 = 5$$

$$u_3 + v_1 = 9$$

$$u_3 + 5 = 9 \ \Rightarrow u_3 = 4$$

$$u_3 + v_2 = 7$$

$$4 + v_2 = 7 \ \Rightarrow v_2 = 3$$

$$u_2 + v_2 = 4$$

$$u_2 + 3 = 4 \ \Rightarrow u_2 = 1$$

$$u_2 + v_3 = 3$$

$$1 + v_3 = 3 \ \Rightarrow v_3 = 2$$

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| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| 12 | $4 - (0 + 3) = 1$ | Increase |
| 13 | $3 - (0 + 2) = 1$ | Increase |
| 21 | $8 - (1+5) = 2$ | Increase |
| 33 | $5 - (4 + 2) = -1$ | Decrease |

| $u_i$ \ $v_j$ | | $v_1$ = 1 | $v_2$ = 2 | $v_3$ = 3 | Supply | |
|---|---|---|---|---|---|---|
| $u_1$ | 1 | 100 [5] | × [4] | × [3] | 100 | 0 |
| $u_2$ | 2 | × [8] | 200+d [4] | 100–d [3] | 300 | 2 |
| $u_3$ | 3 | 200 [9] | × [7] | 100+d [5] | 300 | 4 |
| | Demand | 300 | 200 | 200 | 700 | |
| | | 5 | 2 | 1 | | |

Total cost = 100 × 5 + (200+d) × 4 + (100–d) × 3 + 200 × 9 +(100+d) ×5

= 3900 + 6d

Initially we take $u_1 = 0$

$u_1 + v_1 = 5$

$\because\ u_1 = 0 \qquad 0 + v_1 = 5 \ \Rightarrow v_1 = 5$

$u_3 + v_1 = 9$

$u_3 + 5 = 9 \ \Rightarrow u_3 = 4$

$u_3 + v_3 = 5$

$4 + v_3 = 5 \ \Rightarrow v_3 = 1$

$u_2 + v_3 = 3$

$u_2 + 1 = 3 \ \Rightarrow u_2 = 2$

$u_2 + v_2 = 4$

$2 + v_2 = 4 \ \Rightarrow v_2 = 2$

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| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| 12 | $4 - (0 + 2) = 2$ | Increase |
| 13 | $3 - (0 + 1) = 2$ | Increase |
| 21 | $8 - (2+5) = 1$ | Increase |
| 32 | $7 - (4 + 2) = 1$ | Increase |

Since all the values of occupied cells are non-negative. Hence the solution obtained is optimal.

**Question:** For what value of $S_4$ the problem is unbalance.

$S_1 = 20$ , $S_2 = 30$ , $S_3 = 25$ , $S_4 = ?$

$D_1 = 20$ , $D_2 = 25$ , $D_3 = 40$

**Solution:** First we check at what value of $S_4$ our problem is balance.

$S_4 = 10$

Unbalance for value of $S_4$ is

$0 \le S_4 < 10$

$S_4 > 10$

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## Maximization in transportation problem:

There are certain types of transportation problem where the objective is to be maximized instead of minimized. These problems can be solved by converting the maximization problem into a minimization problem.

**Example:** Find maximum solution of the following transportation problem.

| | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| X | $\lceil 12 \rceil$ | $\lceil 18 \rceil$ | $\lceil 6 \rceil$ | $\lceil 25 \rceil$ | 200 |
| Y | $\lceil 8 \rceil$ | $\lceil 7 \rceil$ | $\lceil 10 \rceil$ | $\lceil 18 \rceil$ | 500 |
| Z | $\lceil 14 \rceil$ | $\lceil 3 \rceil$ | $\lceil 11 \rceil$ | $\lceil 20 \rceil$ | 300 |
| Demand | 180 | 320 | 100 | 400 | 1000 |

**Solution:** Maximization transportation problem can be converted into minimization transportation problem by subtracting each transportation cost from maximum transportation cost. Here the maximum transportation cost is 25. So subtract each value from 25. The revised transportation problem is shown below.

| | 1 | 2 | 3 | 4 | Supply |
|---|---|---|---|---|---|
| X | × 13 | × 7 | × 19 | 200 0 | ~~200~~ × |
| Y | 80 17 | 320 18 | 100 15 | × 7 | ~~500 400 320~~ × |
| Z | 100 11 | × 22 | × 14 | 200 5 | ~~300 100~~ × |
| Demand | ~~180 80~~ × | ~~320~~ × | ~~100~~ × | ~~400 200~~ × | 1000 |

Total cost = 200 × 25 +80 × 8+320×7 +100 × 10 +100 × 14 + 200 × 20 = 14280

To check 14280 is maximum by u-v mehtod.

| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | Supply | |
| $u_1$ | X | × 13 | × 7 | × 19 | 200 0 | 200 | 0 |
| $u_2$ | Y | 80 17 | 320 18 | 100 15 | × 7 | 500 | 11 |
| $u_3$ | Z | 100 11 | × 22 | × 14 | 200 5 | 300 | 5 |
| | Demand | 180 | 320 | 100 | 400 | 1000 | |
| | | 6 | 7 | 4 | 0 | | |

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| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| X1 | 13 – ( 0 + 6) = 7 | Increase |
| X2 | 7 – (0 + 7) = 0 | N.I.N.D |
| X3 | 19 – (0+4) = 15 | Increase |
| Y4 | 7 – (11 + 0) = – 4 | Decrease |
| Z2 | 22–(5+7) = 10 | Increase |
| Z3 | 14–(5+4) = 5 | Increase |

| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | Supply | |
| $u_1$ | X | × 13 | × 7 | × 19 | 200 0 | 200 | 0 |
| $u_2$ | Y | ×17 | 320 18 | 100 15 | 80 7 | 500 | 7 |
| $u_3$ | Z | 180 11 | × 22 | × 14 | 120 5 | 300 | 5 |
| | Demand | 180 | 320 | 100 | 400 | 1000 | |
| | | 6 | 11 | 8 | 0 | | |

Total cost = 200×25+320×7+100×10+80×18+180×14+120×20 = 14600

| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| X1 | 13 – ( 0 + 6) = 7 | Increase |
| X2 | 7 – (0 + 11) = –4 | Decrease |
| X3 | 19 – (0+8) = 11 | Increase |
| Y1 | 17 – (7 + 6) = 4 | Increase |
| Z2 | 22–(11+5) = 6 | Increase |
| Z3 | 14–(5+8) = 1 | Increase |

| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | Supply | |
| $u_1$ | X | × 13 | 200 7 | × 19 | × 0 | 200 | 0 |
| $u_2$ | Y | × 17 | 120 18 | 100 15 | 280 7 | 500 | 11 |
| $u_3$ | Z | 180 11 | × 22 | × 14 | 120 5 | 300 | 9 |
| | Demand | 180 | 320 | 100 | 400 | 1000 | |
| | | 2 | 7 | 4 | –4 | | |

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Total cost = 200×18+120×7+100×10+280×18+180×14+120×20 = 15400

| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| X1 | 13 – ( 0 + 2) = 11 | Increase |
| X3 | 19 – (0 + 4) = 15 | Increase |
| X4 | 0 – (0 –4) = 4 | Increase |
| Y1 | 17 – (11 + 2) = 4 | Increase |
| Z2 | 22–(9+7) = 6 | Increase |
| Z3 | 14–(9+4) = 1 | Increase |

Since all the unoccupied cell are non-negative. So, the solution obtianed is optimal.

**Question:** Find maximum solution by NWCM , also by u-v method.

**Solution:**

| | A | B | C | D | Supply |
|---|---|---|---|---|---|
| 1 | $\boxed{4}$ | $\boxed{6}$ | $\boxed{8}$ | $\boxed{6}$ | 700 |
| 2 | $\boxed{3}$ | $\boxed{5}$ | $\boxed{2}$ | $\boxed{5}$ | 400 |
| 3 | $\boxed{3}$ | $\boxed{9}$ | $\boxed{6}$ | $\boxed{5}$ | 600 |
| Demand | 400 | 450 | 350 | 500 | 1000 |

Maximum value is 9

| | A | B | C | D | Supply |
|---|---|---|---|---|---|
| 1 | 400 $\boxed{5}$ | 300 $\boxed{3}$ | $\times\boxed{1}$ | $\times\boxed{3}$ | ~~700 300~~ × |
| 2 | $\times\boxed{6}$ | 150 $\boxed{4}$ | 250 $\boxed{7}$ | $\times\boxed{4}$ | ~~400 250~~ × |
| 3 | $\times\boxed{6}$ | $\times\boxed{0}$ | 100 $\boxed{3}$ | 500 $\boxed{4}$ | ~~600 500~~ × |
| Demand | ~~400~~ | ~~450 150~~ | ~~350 100~~ | ~~500~~ | 1700 |

Total cost = 400×4+300×6+150×5+250×2+100×6+500×5 = 7750

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Now by u-v method

| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | A | B | C | D | Supply | |
| $u_1$ | 1 | 400 $\boxed{5}$ | 300 $\boxed{3}$ | $\times\boxed{1}$ | $\times\boxed{3}$ | 700 | 0 |
| $u_2$ | 2 | $\times\boxed{6}$ | 150 $\boxed{4}$ | 250 $\boxed{7}$ | $\times\boxed{4}$ | 400 | 1 |
| $u_3$ | 3 | $\times\boxed{6}$ | $\times\boxed{0}$ | 100 $\boxed{3}$ | 500 $\boxed{4}$ | 600 | −3 |
| | Demand | 400 | 450 | 350 | 500 | 1700 | |
| | | 5 | 3 | 6 | 7 | | |

| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| C1 | 1 – ( 0 + 6) = –5 | Decrease |
| D1 | 3 – (0 + 7) = –4 | Decrease |
| A2 | 6 – (1+5) = 0 | N.I.N.D |
| D2 | 4 – (1 + 7) = –4 | Decrease |
| A3 | 6–(–3+5) = 4 | Increase |
| B3 | 0–(–3+3) = 0 | N.I.N.D |

| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | A | B | C | D | Supply | |
| $u_1$ | 1 | 400 [5] | 50 [3] | 250 [1] | × [3] | 700 | 0 |
| $u_2$ | 2 | × [6] | 400 [4] | × [7] | × [4] | 400 | 1 |
| $u_3$ | 3 | × [6] | × [0] | 100 [3] | 500 [4] | 600 | 2 |
| | Demand | 400 | 450 | 350 | 500 | 1700 | |
| | | 5 | 3 | 1 | 2 | | |

Total cost = 400×4+50×6+250×8+400×5+100×6+500×5 = 9000

| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| D1 | 3 – ( 0 + 2) = 1 | Increase |
| A2 | 6 – (1 + 5) = 0 | N.I.N.D |
| C2 | 7 – (1+1) = 5 | Increase |
| D2 | 4 – (1 + 2) = 1 | Increase |
| A3 | 6–(2+5) = –1 | Decrease |
| B3 | 0–(2+3) = –5 | Decrease |

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| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | A | B | C | D | Supply | |
| $u_1$ | 1 | 400 [5] | × [3] | 300 [1] | × [3] | 700 | 0 |
| $u_2$ | 2 | × [6] | 400 [4] | × [7] | × [4] | 400 | 6 |
| $u_3$ | 3 | × [6] | 50 [0] | 50 [3] | 500 [4] | 600 | 2 |
| | Demand | 400 | 450 | 350 | 500 | 1700 | |
| | | 5 | –2 | 1 | 2 | | |

Total cost = 400×4+300×8+400×5+50×9+50×6+500×5 = 9250

| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| B1 | 3 – ( 0 – 2) = 5 | Increase |
| A2 | 6 – (6 + 5) = –5 | Decrease |
| C2 | 7 – (6+1) = 0 | N.I.N.D |
| D2 | 4 – (6 + 2) = –4 | Decrease |
| A3 | 6–(2+5) = –1 | Decrease |
| D1 | 3–(0+2) = 1 | Increase |

| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | A | B | C | D | Supply | |
| $u_1$ | 1 | 350 $\boxed{5}$ | $\times\boxed{3}$ | 350 $\boxed{1}$ | $\times\boxed{3}$ | 700 | 0 |
| $u_2$ | 2 | 50 $\boxed{6}$ | 350 $\boxed{4}$ | $\times\boxed{7}$ | $\times\boxed{4}$ | 400 | 1 |
| $u_3$ | 3 | $\times\boxed{6}$ | 100 $\boxed{0}$ | $\times\boxed{3}$ | 500 $\boxed{4}$ | 600 | –3 |
| | Demand | 400 | 450 | 350 | 500 | 1700 | |
| | | 5 | 3 | 1 | 7 | | |

Total cost = 350×4+350×8+50×3+350×5+100×9+500×5 = 9500

| Unoccupied Cell | $c_{ij} - (u_i + v_j)$ Opportunity cost | Remarks |
|---|---|---|
| B1 | 3 – ( 0+3) = 0 | N.I.N.D |
| D1 | 3 – (0 + 7) = –4 | Decrease |
| C2 | 7 – (1+1) = 5 | Increase |
| D2 | 4 – (1 + 7) = –4 | Decrease |
| A3 | 6–(–3+5) = 4 | Increase |
| C3 | 3–(–3+1) = 5 **By-Pin2** | Increase |

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| | | $v_1$ | $v_2$ | $v_3$ | $v_4$ | | |
|---|---|---|---|---|---|---|---|
| | | A | B | C | D | Supply | |
| $u_1$ | 1 | 350 $\boxed{5}$ | $\times\boxed{3}$ | 350 $\boxed{1}$ | $\times\boxed{3}$ | 700 | 0 |
| $u_2$ | 2 | 50 $\boxed{6}$ | $\times\boxed{4}$ | $\times\boxed{7}$ | 350 $\boxed{4}$ | 400 | 1 |
| $u_3$ | 3 | $\times\boxed{6}$ | 450 $\boxed{0}$ | $\times\boxed{3}$ | 150 $\boxed{4}$ | 600 | 1 |
| | Demand | 400 | 450 | 350 | 500 | 1700 | |
| | | 5 | –1 | 1 | 3 | | |

Total cost = 350×4+350×8+50×3+350×5+450×9+150×5 = 10900

| Unoccupied Cell | $c_{ij} - \left(u_i + v_j\right)$ Opportunity cost | Remarks |
|---|---|---|
| B1 | $3 - (0-1) = 4$ | Increase |
| D1 | $3 - (0 + 3) = 0$ | N.I.N.D |
| B2 | $4 - (1-1) = 4$ | Increase |
| C2 | $7 - (1 + 1) = 5$ | Increase |
| A3 | $6-(1+5) = 0$ | N.I.N.D |
| C3 | $3-(1+1) = 1$ | Increase |

Since all the unoccupied cell are non-negative. So, the solution obtained is optimal.

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Lecture # 12

## Assignment Problem:

The assignment problem is a special type of transportation problem where the objective is to minimize the cost or maximize the profit of the given problem.

## Assumption in Assignment problem:

(i) Number of jobs = Number of Machine or person
(ii) Each person or machine is assigned only one job
(iii) Each person is independently capable of handling any job to be done.
(iv) Assigning criteria is clearly specified (minimizing cost or maximizing profit).

We use Hungarian method to solve assignment problem.

**Question:** What is feasible solution in Hungarian method?

**Answer:** If there are 'n' number of jobs given to 'n' different person's then it is called feasible solution in Hungarian method.

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**Example:** Minimization

J = Jobs

P = Persons

| | $P_1$ | $P_2$ | $P_3$ | $P_4$ |
|---|---|---|---|---|
| $J_1$ | 5 | 9 | 3 | 6 |
| $J_2$ | 8 | 7 | 8 | 2 |
| $J_3$ | 6 | 10 | 12 | 7 |
| $J_4$ | 3 | 10 | 8 | 6 |

## Procedure:

**I:** Identify the minimum value in each row and subtract it from every value in that row.

| | | | |
|---|---|---|---|
| 2 | 6 | 0 | 3 |
| 6 | 5 | 6 | 0 |
| 0 | 4 | 6 | 1 |
| 0 | 7 | 5 | 3 |

**II:** Identify the minimum value in each column and subtract it from every value in that column.

| | | | |
|---|---|---|---|
| 2 | 2 | 0 | 3 |
| 6 | 1 | 6 | 0 |
| 0 | 0 | 6 | 1 |
| 0 | 3 | 5 | 3 |

**III:** If any row or any column has only one zero then make an assignment there and move to next column or row.

| | | | |
|---|---|---|---|
| 2 | 2 | [0] | 3 |
| 6 | 1 | 6 | [0] |
| X | [0] | 6 | 1 |
| [0] | 3 | 5 | 3 |

Where [0] = job assigned

X = job not assigned

Minimum solution = 3+2+10+3 = 18

**Example:** Find minimize solution of the following assignment problem by Hungarian method.

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| | | | | |
|---|---|---|---|---|
| 11 | 7 | 10 | 17 | 10 |
| 13 | 21 | 7 | 11 | 13 |
| 13 | 13 | 15 | 13 | 14 |
| 18 | 10 | 13 | 16 | 14 |
| 12 | 8 | 16 | 19 | 10 |

| | | | | |
|---|---|---|---|---|
| 4 | 0 | 3 | 10 | 3 |
| 6 | 14 | 0 | 4 | 6 |
| 0 | 0 | 2 | 0 | 1 |
| 8 | 0 | 3 | 6 | 4 |
| 4 | 0 | 8 | 11 | 2 |

| | | | | |
|---|---|---|---|---|
| 4 | 0 | 3 | 10 | 2 |
| 6 | 14 | 0 | 4 | 5 |
| 0 | 0 | 2 | 0 | 0 |
| 8 | 0 | 3 | 6 | 3 |
| 4 | 0 | 8 | 11 | 1 |

| | | | | |
|---|---|---|---|---|
| 4 | [0] | 3 | 10 | 2 |
| 6 | 14 | [0] | 4 | 5 |
| [0] | ~~0~~ | 2 | ~~0~~ | ~~0~~ |
| 8 | ~~0~~ | 3 | 6 | 3 |
| 4 | ~~0~~ | 8 | 11 | 1 |

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Here we cannot proceed further because all the zeros are assigned or crossed.

Also $5 \neq 3$

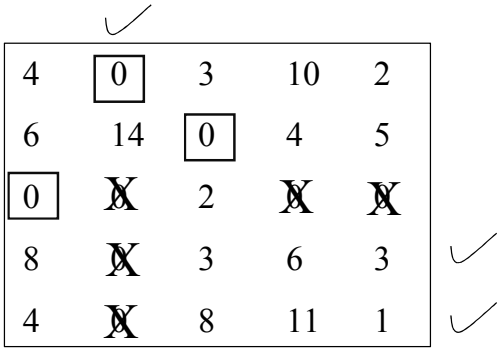
So, this is not feasible. How do we get the other assignments??

We follow the following procedure.

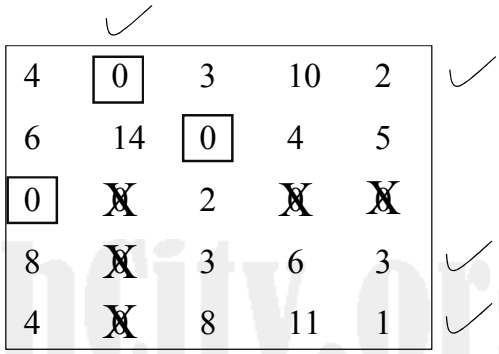
**IV:** (i) Tick all the unassigned rows.

| | | | | | |
|---|---|---|---|---|---|
| 4 | [0] | 3 | 10 | 2 | |
| 6 | 14 | [0] | 4 | 5 | |
| [0] | ~~0~~ | 2 | ~~0~~ | ~~0~~ | |
| 8 | ~~0~~ | 3 | 6 | 3 | ✓ |
| 4 | ~~0~~ | 8 | 11 | 1 | ✓ |

(ii) If a ticked row has a zero then tick the corresponding column (If the column is not yet ticked).



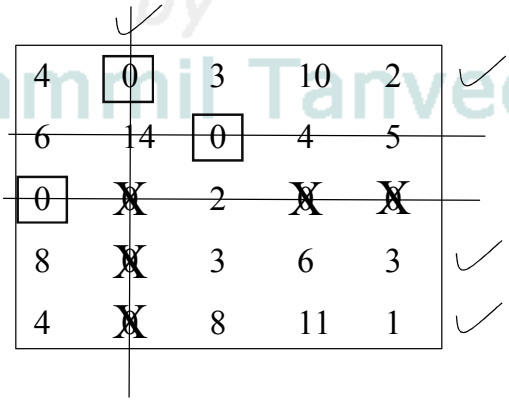

(iii) If a ticked column has an assignment then tick the corresponding row (If the row is not yet ticked).




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(iv) Repeat steps (ii) and (iii) until no more ticking is possible.

(v) Draw lines through unticked rows and ticked column. Number of line represent the number of possible assignment.




(vi) Find out the smallest value which does not have any line passing through and called it θ. $\Rightarrow \theta = 1$

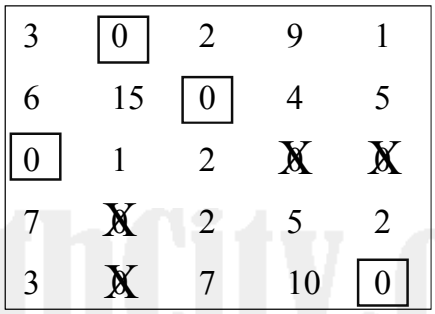
(vii) Add θ if two line is passing through.

(viii) Subtract θ if no line passing through.

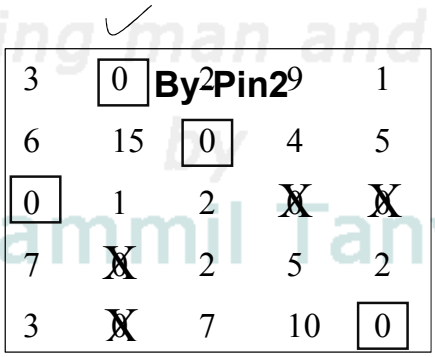
(ix) No change if the value has only one line.

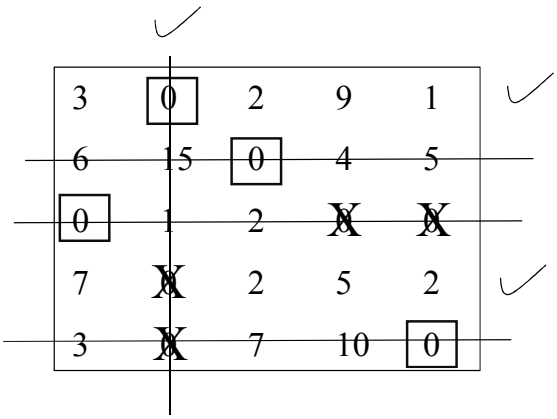
| | | | | |
|---|---|---|---|---|
| 3 | 0 | 2 | 9 | 1 |
| 6 | 15 | 0 | 4 | 5 |
| 0 | 1 | 2 | 0 | 0 |
| 7 | 0 | 2 | 5 | 2 |
| 3 | 0 | 7 | 10 | 0 |

(x) Repeat the above steps again.



$5 \neq 4$





$\theta = 1$

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| | | | | |
|---|---|---|---|---|
| 2 | ~~0~~ | 1 | 8 | ~~0~~ |
| 6 | 16 | [0] | 4 | 5 |
| [0] | 2 | 2 | ~~0~~ | ~~0~~ |
| 6 | [0] | 1 | 4 | 1 |
| 3 | 1 | 7 | 10 | [0] |

$5 \neq 4$

| | | | | |
|---|---|---|---|---|
| 2 | ~~0~~ | 1 | 8 | ~~0~~ |
| 6 | 16 | [0] | 4 | 5 |
| [0] | 2 | 2 | ~~0~~ | ~~0~~ |
| 6 | [0] | 1 | 4 | 1 |
| 3 | 1 | 7 | 10 | [0] |

$\theta = 1$

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| | | | | |
|---|---|---|---|---|
| 1 | ~~0~~ | ~~0~~ | 7 | ~~0~~ |
| 6 | 17 | [0] | 4 | 6 |
| [0] | 3 | 2 | ~~0~~ | 1 |
| 5 | [0] | ~~0~~ | 3 | 1 |
| 2 | 1 | 6 | 9 | [0] |

$5 \neq 4$

$\theta = 1$

| | | | | |
|---|---|---|---|---|
| [0] | ~~0~~ | ~~0~~ | 6 | ~~0~~ |
| 5 | 17 | [0] | 3 | 6 |
| ~~0~~ | 4 | 3 | [0] | 2 |
| 4 | [0] | ~~0~~ | 2 | 1 |
| 1 | 1 | 6 | 8 | [0] |

$5 = 5$

Jobs = Person

Minimum solution = 11+7+13+10+10 = 51

Lecture # 13

**Question:** Find maximum solution of the following Assignment problem.

| | | | | |
|---|---|---|---|---|
| 30 | 37 | 40 | 28 | 40 |
| 40 | 24 | 27 | 21 | 36 |
| 40 | 32 | 33 | 30 | 35 |
| 25 | 38 | 40 | 36 | 36 |
| 29 | 62 | 41 | 34 | 39 |

**Solution:** Here the highest value is 62. So, we subtract each value from it.

| | | | | |
|---|---|---|---|---|
| 32 | 25 | 22 | 34 | 22 |
| 22 | 38 | 35 | 41 | 26 |
| 22 | 30 | 29 | 32 | 27 |
| 37 | 24 | 22 | 26 | 26 |
| 33 | 0 | 21 | 28 | 23 |

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| | | | | |
|---|---|---|---|---|
| 10 | 3 | 0 | 12 | 0 |
| 0 | 16 | 13 | 19 | 4 |
| 0 | 8 | 7 | 10 | 5 |
| 15 | 2 | 0 | 4 | 4 |
| 33 | 0 | 21 | 28 | 23 |

| | | | | |
|---|---|---|---|---|
| 10 | 3 | 0 | 8 | 0 |
| 0 | 16 | 13 | 15 | 4 |
| 0 | 8 | 7 | 6 | 5 |
| 15 | 2 | 0 | 0 | 4 |
| 33 | 0 | 21 | 24 | 23 |

| | | | | |
|---|---|---|---|---|
| 14 | 3 | [0] | 8 | ~~0~~ |
| ~~0~~ | 12 | 9 | 11 | [0] |
| [0] | 4 | 3 | 2 | 1 |
| 19 | 2 | ~~0~~ | [0] | 4 |
| 37 | [0] | 21 | 24 | 23 |

Maximum solution = 40+36+40+36+62 = 214

## Unbalance Problem in Assignment Problem:

**Question:** Minimize

| | | | |
|---|---|---|---|
| 5 | 9 | 3 | 6 |
| 8 | 7 | 8 | 2 |
| 6 | 10 | 12 | 7 |

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**Solution:** 3 Jobs ≠ 4 Person

| | | | |
|---|---|---|---|
| 5 | 9 | 3 | 6 |
| 8 | 7 | 8 | 2 |
| 6 | 10 | 12 | 7 |
| 0 | 0 | 0 | 0 |

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| | | | |
|---|---|---|---|
| 2 | 6 | [0] | 3 |
| 6 | 5 | 6 | [0] |
| [0] | 4 | 6 | 1 |
| ~~0~~ | [0] | ~~0~~ | ~~0~~ |

Minimum solution = 3+2+6+0 = 11

Also find maximum solution.

Here the maximum value is 12. So, we subtract each value from it.

| | | | |
|---|---|---|---|
| 7 | 3 | 9 | 6 |
| 4 | 5 | 4 | 10 |
| 6 | 2 | 0 | 5 |
| 12 | 12 | 12 | 12 |

| | | | |
|---|---|---|---|
| 4 | [0] | 6 | 3 |
| [0] | 1 | ~~0~~ | 6 |
| 6 | 2 | [0] | 5 |
| ~~0~~ | ~~0~~ | ~~0~~ | [0] |

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Maximum solution = 9+8+12+0 =29

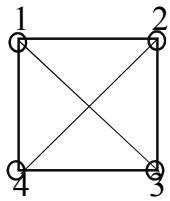
If the question arise find the difference between maximum and minimum solution of Assignment problem then we find both maximum and minimum solution and then subtract minimum solution by maximum solution.

## Network Minimization:

Network minimization deals with the determination of the branches that can join all the vertex of a network, such that lenghts of the choosen branches are minimized. This minimum Network is called a minimal spanning tree.

## Graph/Network:

A graph is a pair G(V,E) where V is the set of vertices/nodes and E is a set of Edges/Branches. Number of vertex in a graph is called order of the graph and number of edges in a graph is called size of graph.

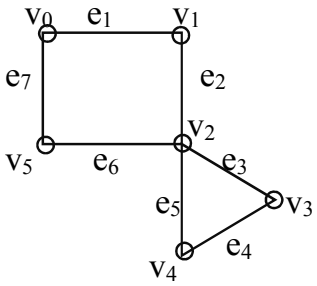



V = {1,2,3,4}  E ={12,23,34,24,31,41}

**Walk:** A walk from u to v in a graph G is a finite alternative sequence of vertices and edges. In a walk vertices and edges can repeat.

**Trail:** A trail is a walk in which no edge can repeat.

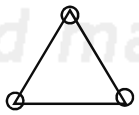
**Path:** A path is a walk in which no vertex and no edge can repeat.




| | | |
|---|---|---|
| I. | $v_0e_1v_1e_2v_2e_6v_5$ | Path |
| II. | $v_0e_1v_1e_2v_2e_3v_3e_4v_4e_5v_2$ | Trail |
| III. | $v_3e_4v_4e_5v_2\ e_3v_3e_4v_4$ | Walk |

**Cylce:** A closed path is called a cycle.

*smallest cycle is $C_3$ **By-Pin2**



**Connected graph:**

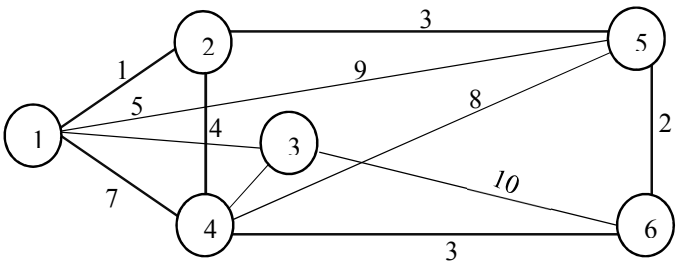
A non-emtpy graph G is connected if for any two vertices we have a path.

**Tree:** An acyclic (in which no cycle) connected graph is called tree.

**Spanning graph:** A spanning graph of a graph G that includes all the vertices of graph G.

**Minimal spanning tree:** A tree T is called minimal spanning tree if T is connected then T−e is disconnected for all e ∈ E.

**Question:** Find minimal spanning tree of the following network.



**Solution:** $I_1$ = start with node 1

$C_1 = \{1\}$ , $\overline{C_1} = \{2,3,4,5,6\}$

$I_2$ = connect node 2 with node 1

$C_2 = \{1,2\}, \overline{C_1} = \{3,4,5,6\}$

$I_3$ = connect node 5 with node 2

$C_3 = \{1,2,5\}, \overline{C_3} = \{3,4,6\}$

$I_4$ = connect node 6 with node 5

$C_4 = \{1,2,5,6\}, \overline{C_4} = \{3,4\}$

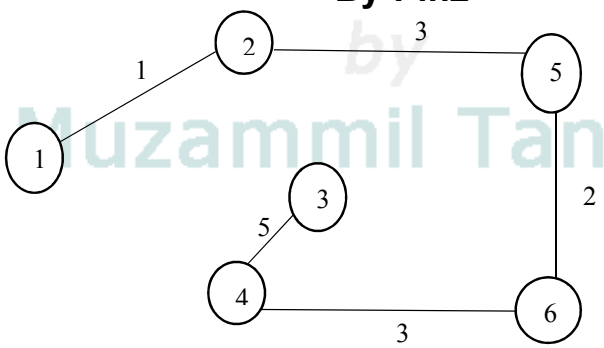
$I_5$ = connect node 4 with node 6

$C_5 = \{1,2,4,5,6\}, \overline{C_5} = \{3\}$

$I_6$ = connect node 3 with node 4

$C_6 = \{1,2,3,4,5,6\}, \overline{C_6} = \{\ \}$

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Minimum distance = 1+3+2+3+5 = 14